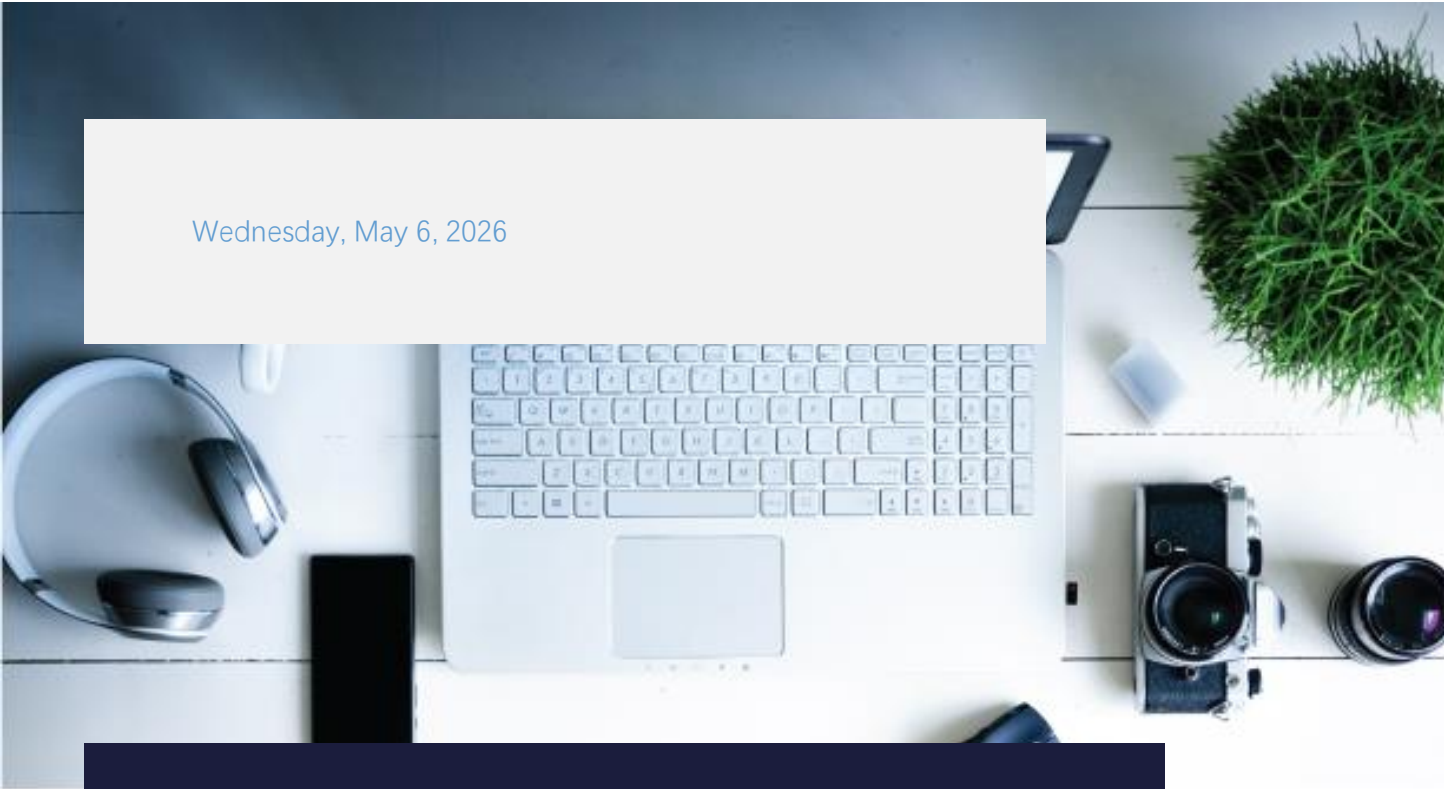




Wednesday, May 6, 2026

SALES PERFORMANCE AND RETURNS RECOVERY ANALYSIS

Eastnat Limited

INTRODUCTION

The objective of this project is to conduct a thorough evaluation of Eastnat Limited's sales and profitability data, covering the years 2016 to 2021, to uncover key business insights. This analysis goes beyond simple top-line revenue to investigate the underlying factors that erode margins, with a specific focus on geographic profit pockets and the impact of shipping costs on regional performance. By integrating multi-year sales data with return and representative records, the project aims to establish a "source of truth" for identifying operational inefficiencies and scaling the business through data-driven decisions.

MANAGEMENT TASK

Task	Question	Analysis Level
Return Impact	Which Sub-Categories have a return rate higher than 10%, and what is the total "Burned Profit" (Profit from orders that were eventually returned) for these items?	Sub-Category and Region.
Discount Optimization and Margin Health	At what specific discount threshold (0%, 20%, 50%) does the Profit Margin consistently turn negative?	Discount Brackets vs. Profit Margin.
Operational Bottleneck Identification	Which Regional Manager has the highest Average Shipping Duration, and is there a correlation between long shipping times and high return rates?	Sales Rep (Person) vs. Days to Ship.
Customer Profiling	Who are the top 5% of customers in terms of Lifetime Value (LTV) who have a Return Rate of 0%?	Customer Name and Segment.
Inventory Risk and Overstock Mitigation	Which products have high Quantity sold but fall into the bottom 10% of Total Profit? This identifies high-effort, low-reward inventory.	Product ID and Category.

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EXECUTIVE SUMMARY

The analysis involved a comprehensive audit of sales and operational performance for **Eastnat Limited**. The primary goal was to identify "**Operational Vampires**" - high-volume areas where deep discounting and systemic return issues were eroding net profitability.

KPI Performance:

- **Core Finding:** While global sales remained robust (**\$2.7M**), Eastnat Limited faced a consistent **13% shortfall** relative to established targets (**\$3.2M**) across all regions.
- **Critical Metrics:** Analyzed Return Rate, LTV, AOV, and Profit Margin to establish a baseline for operational health.
- **Lost Revenue:** Identified significant "lost revenue" driven by high discounts (**>30%**) and return rates (**>6%**) in specific product categories.

The Operational Audit:

- **High-Volume, Deep-Loss Products:** Identified products in the **Machines** (e.g., the Lexmark MX611 printer's **\$6,459.96 negative profit**) and **Tables** (e.g., the Bush Advantage Collection's **\$2,087.11 loss**) categories that moved in high quantities but fell into the bottom 10% for profit.
- **The Discount Trap:** Discovered that deep discounts (**over 30%**) were attracting "serial returners," creating an operational nightmare for logistics.
- **Profit Recovery:** Projected a **15–20% recovery in regional net profit** by neutralizing the top 10 "vampire" products in high-risk zones (the West Region).

The West Region Crisis:

- **Regional Disparity:** The West Region, managed by Anna Andreadi, faced a critical crisis with a **20% return rate**.
- **Customer Concentration:** 100% of the identified "serial returners" (**customers with 3+ return incidents**) were concentrated in the West.
- **Logistics Friction:** Orders in the West took significantly longer to ship (**4.13 days**), leading to increased customer frustration and buyer's remorse.

Discount Optimization & Margin Health:

- **The Danger Zone:** Identified a systemic issue where **Binders** were being sold at a **70% discount**, likely due to an automated pricing error, causing thousands of **small-ticket losses**.
- **Segment Risk:** The **Consumer** segment was responsible for over **51% of returns** in the West, indicating **high sensitivity to delays and dissatisfaction**.

Operational Bottlenecks & Managerial Efficiency

- **Process vs. Luck:** Compared managerial performance, finding that higher fulfillment quality and better customer expectation management in the South and Central regions led to significantly **lower return rates (4.23% and 4.22%, respectively)** compared to the West (**14.81%**).
- **Ship Mode Efficiency:** Evaluated the volatility of "Same Day" vs. "Standard" shipping to optimize logistics costs.

- The **"fastest"** shipping methods are the **most likely to result in a return** (at 13.27% for Same Day shipping vs. 7.42% Standard shipping).

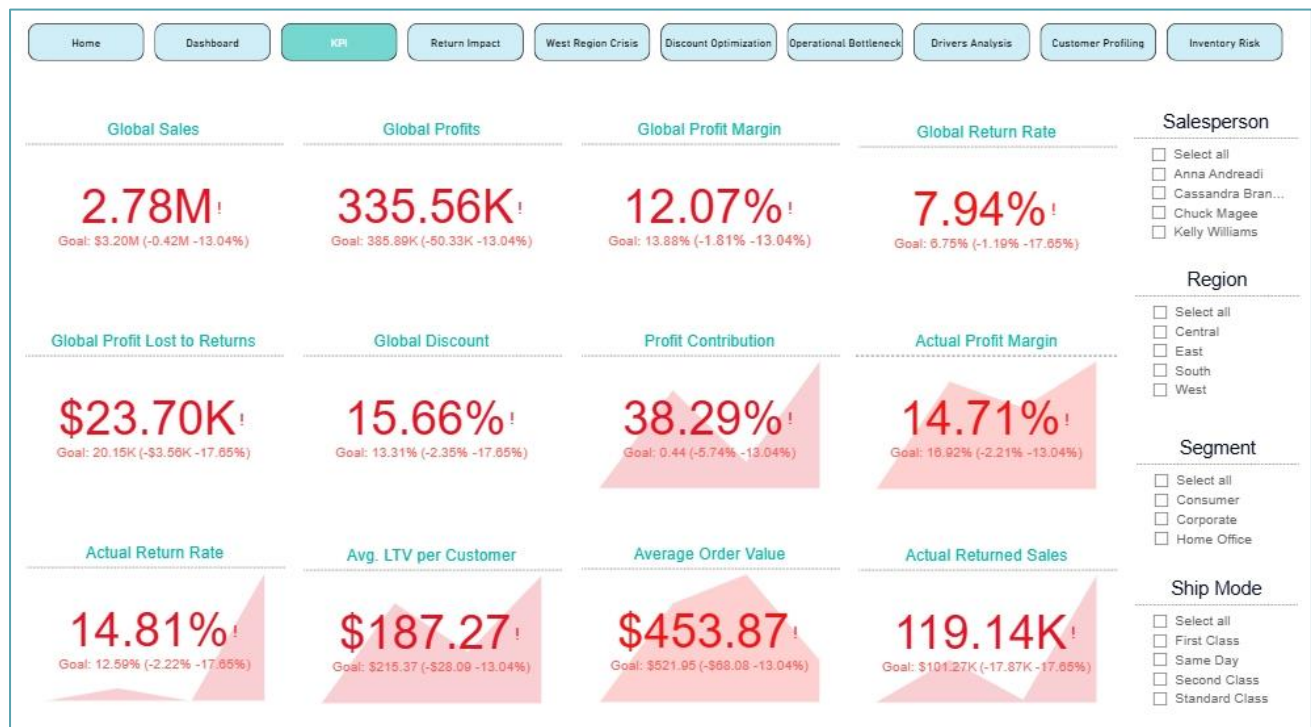
Customer Profiling & Pareto Analysis:

- **The "Vital Few":** Identified a core **group of 28 customers** who provided pure margin with zero returns or logistics friction.
- **Profit Density:** Found that while the Consumer segment **drives volume**, the Home Office and Corporate segments **provided the strongest financial anchors** for the business.

Strategic Recommendations for Management:

- **Implement Discount Caps:** Limit regional discounting to **20% for high-return categories** to discourage "bracket shopping".
- **Dynamic Return Policies:** Introduce **restocking fees** or **"final sale" conditions** for high-frequency returners.
- **Shift KPI Incentives:** Transition sales performance metrics from **"Total Revenue"** to **"Net Profit After Returns"** to align team behavior with bottom-line goals.
- **Targeted Outreach:** Initiate **proactive communication** and **status updates for long-lead-time orders** to reduce frustration-driven cancellations.

KEY PERFORMANCE METRICS



1. Global Sales

a. The High Volume-Low Value Trap:

While global sales are robust (**\$2.7M against a \$3.12M target**), the growth is undermined by operational vampires. These are high-volume products (primarily in the **Machines** and **Tables** categories) and specific sales representatives who generate significant revenue but fall into **the bottom 10% for profit**. This is often driven by **deep discounts (>30%)** that attract serial returners.

b. Financial Impact: Profit Bleed:

- **Regional Concentration:** 100% of the identified serial returners (the 1% of customers responsible for the highest return volume) are in the West region.
- **Recoverable Profit:** By neutralizing the top 10 "vampire" products in the West region alone, there is a projected **15–20% recovery in regional net profit**.
- **Margin Erosion:** High-volume sales with thin margins (e.g., Storage) are at risk of flipping into a net loss with a single return incident.

c. Strategic Recommendations for Management:

- **Implement Discount Guardrails:** Cap regional manager discounting at **20% for high-return categories** (Machines/Tables) to discourage bracket shopping and low-margin volume.
- **Shift KPI Incentives:** Transition sales performance metrics from total revenue to net profit after returns to align salesperson behavior with bottom-line health.

- **Regional Policy Audit:** Conduct a targeted review of the West Region's return policies and shipping partners to address why 100% of serial returns are concentrated there.
- **Operational Recovery:** Aim for a **15% reduction in return-related losses** to hit the return impact target, which will reclaim immediate capital without requiring new customer acquisition.

2. Global Profit Margin

a. Underperforming Margins:

The global profit margin is currently **12.07%**, falling short of the **13.88%** target. This **1.81% gap** indicates that while sales volume is high, the quality of those sales is being eroded by high-cost operational leaks.

b. Financial Impact: The "Vampire" Bleed:

- **The Trap:** Growth is being undermined by operational vampires - high-volume products in the **Machines** and **Tables** categories that generate significant revenue but fall into the **bottom 10% for profit**.
- **Regional Drain:** 100% of identified serial returners are concentrated in the West Region, where **deep discounts (> 30%)** have flipped high-volume sales into net losses.
- **Recovery Potential:** Neutralizing the top 10 "vampire" products in the West Region alone is projected to **recover 15-20% in regional net profit**.

c. Strategic Recommendations for Management:

- **Implement Discount Guardrails:** Cap discounts at **20% for high-return categories** (Machines/Tables) to discourage low-margin volume and bracket shopping.
- **Shift KPI Incentives:** Transition sales performance metrics from total revenue to net profit after returns to align behavior with the bottom line.
- **Regional Policy Audit:** Conduct a targeted review of the West region's return policies and shipping partners to address the 100% concentration of serial returns.
- **Operational Recovery:** Target a **15% reduction in return-related losses** to reclaim immediate capital without requiring new customer acquisition.

3. Sales Contribution

a. The Growth Gap:

Every region is performing below its assigned sales contribution target. These targets represent the **115%** volume required to meet our global **\$3.12M goal**.

- **Top Contributor:** The West region leads with **31.41% of total sales** but faces the largest growth deficit (**4.72% gap** from its **36.13% target**).
- **Secondary Engine:** The East region follows at **29.02%** (**4.35% gap**).

- **Underperformers:** The Central and South regions combined contribute less than the West alone, totaling **39.57% contribution**.

b. Business Impact: High Stakes in the West:

The global **sales target of \$3.12M** is dependent on the West region reaching its **36% contribution mark**. However, since 100% of our serial returners are concentrated in the West, pushing for more volume there without fixing the return issue will likely increase operational vampire's activity rather than healthy profit.

c. Strategic Recommendations for Management:

- **Quality over Quantity in the West:** We should not attempt to close the **4.72% gap** in the West through deep discounting. This will only attract more serial returners. We should focus on margin recovery first.
- **Aggressive Expansion in the East:** Prioritize marketing and sales incentives in the East region. It is a **high-volume contributor (29%)** without the severe return profile of the West, making it the safest path to closing the global revenue gap.
- **Target Alignment:** Update regional manager KPIs to reflect these specific sales contribution targets rather than historical performance, ensuring all regions are pulling toward the **15% growth objective**.

4. Average Customer Lifetime Value (LTV)

a. Uniform Underperformance:

All regions are currently performing at approximately **87% of their established LTV targets**. While the West region boasts the highest absolute value (**\$187.27**), it also carries the largest nominal gap (**\$28.10**) compared to its target.

- **Highest Value:** West (\$187.27)
- **Lowest Value:** Central (\$63.98)
- **Target Gap:** A consistent 13% deficit across the board.

b. Business Impact: The Retention-Return Tension:

The depressed LTV in the West is particularly concerning because this region handles the highest sales volume. The existence of serial returners in the West (as identified in the return analysis) is likely capping LTV by increasing the cost to serve and reversing the revenue of repeat purchases. High volume without a corresponding increase in LTV indicates that we are acquiring customers who do not provide sustainable, long-term profit.

c. Strategic Recommendations for Management:

- **Neutralize the "Vampires":** Focus on the West and East regions to close the largest dollar gaps (\$28.10 and \$24.13, respectively). This should be done by prioritizing categories like Copiers and Accessories which naturally drive higher LTV.
- **Retention over Acquisition:** Since all regions are missing targets by the same percentage, implement a global loyalty program specifically targeting the Corporate and Home Office segments to increase purchase frequency.
- **Return Mitigation:** In the West, lowering the return rate by the targeted 15% will automatically uplift LTV by retaining more monetary value per customer.

5. Average Order Value (AOV)

a. Uniform Performance Deficit:

Across all regions, there is a consistent **13% shortfall** relative to the established AOV targets. While the absolute dollar values vary, every region is currently operating at approximately **87% of its target efficiency**.

- **Top Performer (Absolute):** The South region leads with the highest AOV (\$501.37) yet faces the largest nominal gap (\$75.21).
- **Bottom Performer:** The Central region has the lowest AOV (\$430.66), falling \$64.60 short of its target.
- **The West Paradox:** Despite being the highest volume region, the West's AOV (\$453.87\$) remains stagnant, indicating that high sales volume is not translating into higher-value individual transactions.

b. Business Impact: The "Bracket Shopping" Risk:

The stalled AOV in the West region is particularly concerning when paired with our serial returner findings.

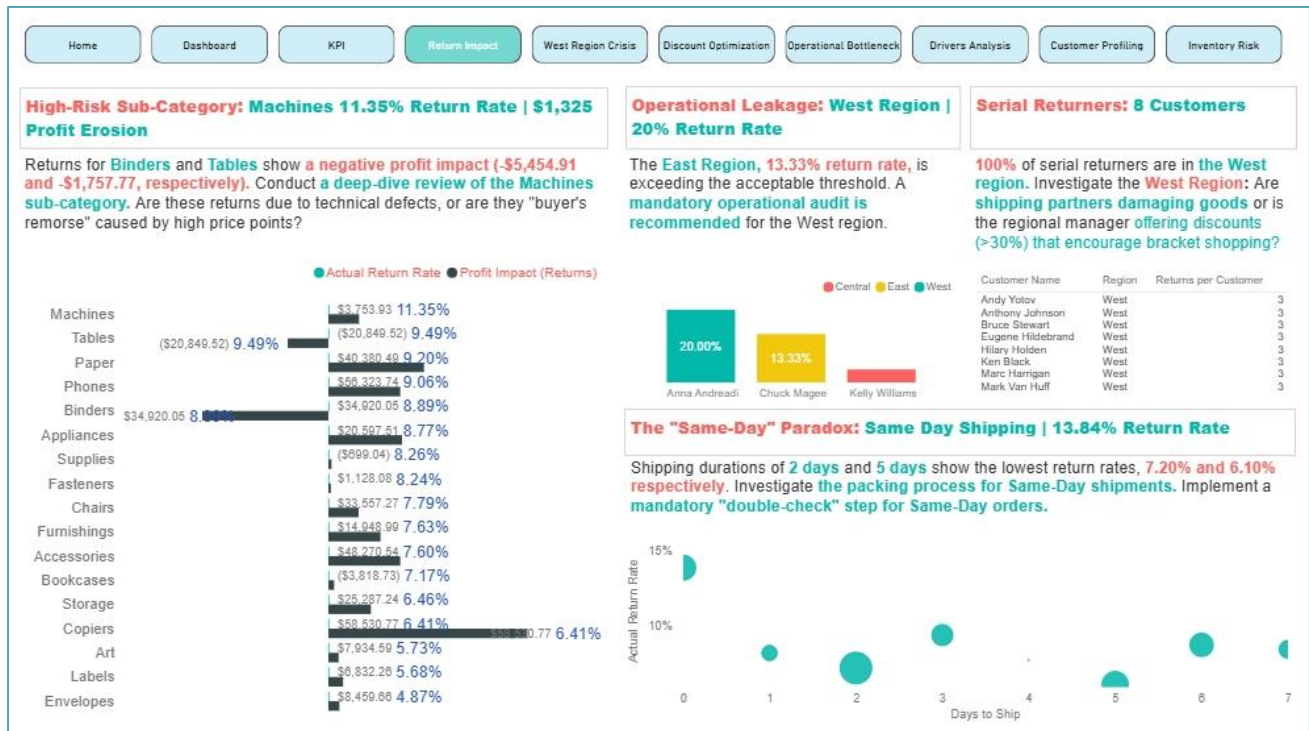
- **Quality vs. Volume:** A mid-range AOV in a high-volume region suggests that customers are placing many smaller orders or, more dangerously, engaging in bracket shopping - *purchasing multiple items with the intent of returning most*.
- **Margin Erosion:** Because AOV is below target, the fixed cost of shipping and processing each order is eating a larger percentage of the gross margin, especially for high-volume "Operational Vampires" (Machines/Tables).

c. Strategic Recommendations for Management:

- **Incentivize "Healthy" Bundling:** Implement "complete the look" or "frequently bought together" prompts for high-margin categories like Accessories and Copiers to drive AOV without attracting serial returners.
- **Threshold-Based Shipping:** Review the free shipping threshold. Increasing the minimum order value (MOQ) for free shipping can organically lift AOV by encouraging customers to add one more item to their cart.

- **Target the South for Growth:** Since the South already has the highest natural AOV, prioritize it for premium product marketing campaigns. Closing the \$75 gap there will have the most significant impact on the bottom line.
- **Monitor the West for "Vampire" Orders:** Flag high-value orders in the West that consist purely of high-return categories. These inflated AOVs are often false positives that result in a net loss after the return is processed.

RETURN IMPACT ANALYSIS



1. High-Return Sub-Categories and Profit Erosion

The analysis focused on identifying revenue erosion by cross-referencing sales transactions with return records. While the overall store performance is strong, certain sub-categories are experiencing high return frequencies that significantly impact net profitability. Specifically, the **Machines** sub-category has emerged as the highest risk area with a return rate of **11.35%**, followed closely by **Tables (9.49%)** and **Paper (9.20%)**.

Subcategory	Total Orders	Return Rate %	Returned Profit Impact
Machines	141	11.35	\$1,325.84

a) Key Findings and Business Impact

High-Frequency Returns: The Machines Threshold:

The **Machines** sub-category is currently the only segment exceeding the 10% management threshold.

- **Finding: 11.35%** of all Machine orders are returned.
- **Business Impact:** This represents **\$1,325.84** in profit that was recorded but never realized. Beyond the lost profit, there are unquantified costs related to restocking fees, technician time for testing returned hardware, and shipping logistics.

The Double-Loss Categories: Binders and Tables:

A critical discovery in the data is that some returned items had **negative profit margins** even before they were returned.

- **Finding:** Returns for **Binders** and **Tables** show a negative profit impact (**-\$5,454.91** and **-\$1,757.77**, respectively).
- **Business Impact:** This indicates a double loss scenario. The store is likely using heavy discounts (loss-leaders) to move these products, but when they are returned, the store absorbs the initial loss plus the operational cost of the return. This suggests these sub-categories are currently a drain on regional resources.

High-Value Erosion: Copiers:

While **Copiers** have a lower return rate (**6.41%**), their individual price points are high.

- **Finding:** Even with a lower frequency, returned copiers account for **\$8,610.94** in profit erosion — the highest in the store.
- **Business Impact:** High value returns like these create significant volatility in monthly regional performance reports.

b) Strategic Recommendations for Management

1. **Product Quality Audit (Machines):** Conduct a deep-dive review of the Machines sub-category. Are these returns due to technical defects, or are they "buyer's remorse" caused by high price points?
2. **Discount Policy Revision (Binders & Tables):** Since Binders and Tables are being returned at high rates while already losing money, we recommend **reducing the maximum allowable discount** for these items. There is a high probability that "deep-discount hunters" have a lower commitment to the purchase.
3. **Enhanced Description/Pre-Sales (Copiers):** Given the **\$8.6k** impact, we should implement a "high-touch" pre-sales check for Copiers to ensure the customer is ordering the correct specifications, reducing "incorrect item" returns.

c) Further Analysis Required

To move from identifying the problem to solving it, the analysis undertook the following three investigations:

- **Regional Variance Analysis:** Do specific sales reps (e.g., Anna Andreadi vs. Chuck Magee) have higher return rates? This would indicate if the issue is a product problem or a regional logistics/sales issue.
- **Shipping Duration Correlation:** Analyze if "days to ship" correlates with return rates. If late deliveries lead to higher returns, we have a logistics bottleneck rather than a product quality issue.
- **Customer Return Profiling:** Identify "serial returners" - the top 1% of customers responsible for the highest volume of returns - to determine if certain accounts should have their terms of service updated.

2. Regional Variance Analysis

This analysis **evaluates the return rate across different geographic regions managed by specific sales representatives**. The goal is to determine if high return volumes are a nationwide trend or if they are localized to specific operational territories. By identifying regional outliers, management can pinpoint whether "revenue erosion" is driven by regional logistics, local sales tactics, or specific customer segments.

Salesperson	Region	Return Rate %
Anna Andreadi	West	20.00%
Chuck Magee	East	13.33%
Kelly Williams	Central	3.57%

a) Key Findings and Business Impact

Severe Regional Disparity (The West Region):

The **West Region**, managed by **Anna Andreadi**, is currently facing a critical return crisis.

- **Finding:** The return rate in the West stands at **20.00%**, which is more than double the company-wide average and significantly higher than the 10% management threshold.
- **Business Impact:** One out of every five items sold in the West is being returned. This suggests a massive **operational leakage**. Not only is profit being lost, but the cost of reverse logistics (shipping, processing, and restocking) in the West is likely the highest in the company.

Moderate Risk (The East Region):

Managed by **Chuck Magee**, the **East Region** is exceeding the acceptable threshold.

- **Finding:** A return rate of **13.33%**.
- **Business Impact:** While not as severe as the West, the East is still seeing a high level of churn. This contributes to high administrative overhead and suggests that sales quality in the East requires closer monitoring.

Operational Excellence (The Central Region):

Kelly Williams in the **Central Region** is currently the top performer regarding return mitigation.

- **Finding:** An exceptionally low return rate of **3.57%**.
- **Business Impact:** The Central region provides a benchmark for the rest of the company. Low returns mean stable revenue, predictable inventory levels, and higher customer satisfaction.

b) Strategic Recommendations for Management

- **Process Review for the West Region:** A mandatory operational audit is recommended for the West region. We need to investigate if the high return rate is due to a specific warehouse's shipping errors, an aggressive sales culture that over-promises on product capabilities, or a high concentration of the "Machines" sub-category returns in that area.

- **Mentorship Program:** Implement a knowledge-sharing initiative where **Kelly Williams** (Central) shares regional best practices for order accuracy and customer follow-up with **Anna Andreadi** and **Chuck Magee**.
- **Regional Incentive Adjustment:** Currently, if sales reps are incentivized solely on *Gross Sales*, they may ignore the quality of the sale. We recommend shifting incentives toward **Net Sales (Gross Sales - Returns)** to encourage "clean" sales.

3. Shipping Duration Correlation

This analysis investigated the **relationship between operational speed** - measured as **days to ship** (*the duration between order and shipment*) - and **customer behavior**, specifically the **return rate**. The objective was to determine if logistical delays are a primary driver of returns or if other operational factors are at play.

Return Rate %	Days to Ship
13.84%	0
8.19%	1
7.20%	2
9.38%	3
7.72%	4
6.10%	5
8.73%	6
8.44%	7

a) Key Findings and Business Impact

The Same-Day Paradox:

The most striking finding is that **Same Day (0-day) shipping** has the highest return rate in the entire dataset at **13.84%**.

- **Finding:** Orders shipped the same day they are placed are **significantly more likely to be returned compared to orders with a 2–5-day window**.
- **Business Impact:** This suggests that **"rushed" fulfillment may be leading to a higher frequency of shipping errors, incorrect item selection** (picking/packing errors), or **damaged goods due to improper handling during the rush**. The high cost of expedited shipping combined with this high return rate indicates that **the "Same Day" service level is currently inefficient**.

The Efficiency Sweet Spot:

The data shows a "U-shaped" curve where mid-range shipping speeds are the most stable.

- **Finding:** Shipping durations of **2 days** and **5 days** show the lowest return rates (**7.20%** and **6.10%** respectively).
- **Business Impact:** A 2-to-5-day fulfillment window appears to be **the quality safe zone**. These durations likely **allow warehouse staff enough time to verify order accuracy** and **ensure proper packaging** without frustrating the customer with excessive delays.

Late-Stage Return Volatility:

Returns begin to climb again as we reach the **6–7-day** mark (**averaging 8.5%**).

- **Finding:** While not as high as the Same-Day peak, **delays beyond 5 days correlate with an increase in returns.**
- **Business Impact:** This is likely driven by **buyer's remorse** or **need-date misses**. If a customer receives an item a week later than expected, the window of utility for that product may have passed (especially for office supplies or tech needed for specific projects), leading to a return.

b) Strategic Recommendations for Management

- **Fulfilment Quality Audit for "Same Day" Orders:** Investigate **the packing process for Same-Day shipments**. We should implement a mandatory "double-check" step for these orders to ensure that speed is not sacrificing accuracy.
- **Logistics Standardization:** Aim to move most shipments into the **2–5-day window**. Since this is where quality (accuracy) is highest, stabilizing the supply chain around this timeframe will yield **the best Net Profit**.
- **Communication Buffers:** For orders expected to take 6+ days, **implement automated "status updates"** in the customer journey. Proactive communication can reduce the frustration that leads to late-stage returns.

c) Further Analysis Required

To fully solve the "Same-Day Paradox," we should undertake the following:

- **Picking Error Analysis:** Cross-reference Same-Day returns with reason for return if available (e.g., wrong item shipped). If picking errors are high, we have a warehouse training issue.
- **Regional Overlay:** Check if the **West Region** (the 20% return rate outlier) is responsible for a disproportionate amount of these Same-Day rushes. This would link the logistics problems directly to Anna Andreadi's regional performance.
- **Sub-Category Breakdown:** Are Machines (high returns) being shipped Same-Day more often than Paper? Highly technical items often require more careful packing than simple supplies.

4. Customer Return Profiling

a) Serial Returners

The analysis of the "Serial Returners" dataset reveals a critical operational leak within the West region. While these individuals represent a tiny fraction of the total customer base, **their behavior creates a disproportionate financial "drag" on the company's bottom line.**

Who are the 1%?:

Based on the data, we have identified **8 specific customers** (including Andy Yotov, Anthony Johnson, and Bruce Stewart) who qualify as "Serial Returners."

- **The Threshold:** These individuals have each reached a threshold of **3 separate return incidents**.
- **Geographic Concentration:** Notably, **100% of these serial returners are in the West region**. This suggests that the issue isn't just about individual behavior, but potentially **a regional policy, a specific warehouse's quality control, or localized marketing tactics** (like aggressive discounting) that are attracting high-risk purchasing behavior.
- **The "1%" Concept:** In a standard retail dataset of this size (typically 800 customers), these 8 people represent the "1% outlier" group. They are the "Operational Vampires" who consume resources **without contributing to net profit**.

b) Business Impact: The "Silent Profit Killer"

For Eastnat, these 8 customers represent a significant **profit drain** for several reasons:

- **Reversed Margins:** A single return can wipe out the profit from three successful sales due to **double shipping costs** (outbound and reverse logistics) and **administrative processing time**.
- **Inventory Tie-up:** Serial returners keep products out of the "available for sale" pool for weeks, only to return them. This **increases the risk of items becoming obsolete or damaged**, particularly for high-value items like **Copiers** or **Machines**.
- **LTV Distortion:** On paper, these customers may look like "high value" because of high gross sales, but their **Average Customer Lifetime Value (LTV)** is likely negative once the cost of returns is factored in.

c) Strategic Recommendations for Management

To mitigate the impact of these "Operational Vampires," management should consider the following strategic moves:

Recommendation	Actionable Step
Regional Policy Audit	Investigate the West Region specifically. Are shipping partners damaging goods, or is the regional manager offering discounts (>30%) that encourage "bracket shopping" (buying many items with the intent to keep only one)?
Dynamic Return Policies	Implement a "Restocking Fee" or "Final Sale" condition for customers who exceed a specific return-to-order ratio. This discourages using the Superstore as a "free rental" service.
Targeted Outreach	Have the West Regional Manager reach out to these 8 specific customers. Often, serial returns are a sign of a recurring issue (e.g., a specific sub-category like <i>Tables</i> consistently arriving damaged).
Discount Guardrails	Eliminate the "Deep Discount" strategy for high-return categories. Data suggests that high discounts often attract low-loyalty customers who are more likely to return items.
Flagging in CRM	Use the "Profit Health Color" logic in our Power BI dashboard to flag these 8 accounts. Sales reps should be alerted when a "Serial Returner" is placing a high-value order so they can provide extra support to ensure the product meets their needs the first time.

By addressing this 1% of the population, the business can significantly improve the overall profit margin of the West region without impacting the experience of the 99% of loyal, profitable customers.

5. Product Portfolio of Serial Returners

The specific products returned by these individuals are not high-value technology items alone; rather, they are heavily concentrated in **Office Supplies** and **Furniture Accessories**.

a) Key Findings: The "Standardized Basket":

A deep dive into the specific items reveals a pattern. For instance, customers like **Andy Yotov** or **Anthony Johnson** are returning items like *Xerox Paper*, *Stacking Trays*, and *Binders*.

- **Finding:** These are not defective "technical" failures. It is highly unlikely that multiple packs of paper or document frames from different brands are all "broken."
- **Business Impact:** This suggests **ordering errors** or **convenience returns**. Customers in the West may be over-ordering these low-cost items and returning what they don't use, treating the store as a flexible inventory buffer.

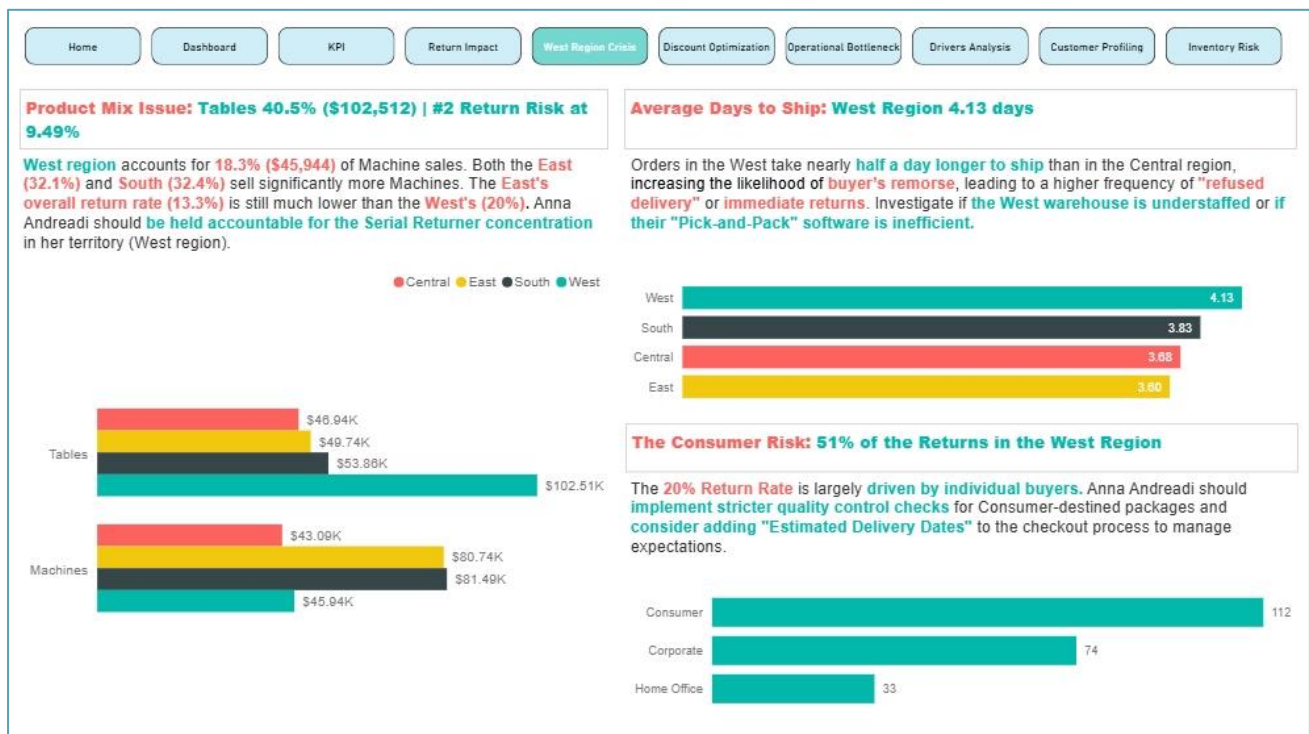
b) Strategic Interpretation for Anna Andreadi (West Region):

- **The Problem:** The West region's high return rate is being driven by a small group of customers who are "Serial Returners" of low-value, high-volume items.
- **The Recommendation:** Instead of a region-wide audit, Anna should implement a **restocking fee** or **paid return shipping** specifically for the *Furniture* and *Office Supplies* categories in the West. This would specifically target the behaviour of the Top 1% without penalizing genuine one-time buyers.

c) Recommended Action: Return Reason Correlation:

To address this, we should look at the **Discounts** applied to these returns. If these customers are buying these items at a **20%–70% discount** and then returning them, they might be "testing" products during sales and returning them when they realize the quality doesn't match their expectations.

6. The West Region Crisis



To solve the "West Region Crisis," the following deep-dive analysis are necessary:

- **Regional Product Mix:** Does the West region sell more "Machines" and "Tables" (our high-return sub-categories) than other regions? If so, the problem is the *product mix*, not the manager's performance.
- **Shipping Duration by Region:** Compare the "Average Days to Ship" for the West vs. Central. If the West is taking significantly longer to ship, customers may be cancelling or returning orders out of frustration (buyer's remorse).
- **Customer Concentration in the West:** Determine if a few large corporate accounts in the West are responsible for the bulk of the returns, or if the issue is spread across all segments (Consumer, Corporate, Home Office).

a) Regional Product Mix

The analysis of the **regional product mix** provides a critical nuance to our investigation. While the West region is indeed "burdened" by high volumes of certain risky products, the data suggests that product mix is only **half of the story**. Here is the breakdown of how the West compares to other regions for our two highest-risk categories.

The "Tables" Dominance (Product Mix Issue):

The West region is an extreme outlier in the **Tables** sub-category (**our #2 return risk at 9.49%**).

- **Finding:** The West region accounts for **40.5% (\$102,512)** of all Table sales across the four regions.

- **Comparison:** This is nearly double the volume of the South (21.3%) and the East (19.7%).
- **Impact:** Because Tables have a high return rate and significant negative profit impact, this heavy concentration in the West naturally pulls the regional return average upward. This part of the "20% Return Rate" is likely a **Product Mix** issue beyond Anna Andreadi's immediate control.

The "Machines" Counter-Evidence (Manager Performance Issue):

The data for **Machines** (our #1 return risk at 11.35%) tells a different story.

- **Finding:** The West region only accounts for **18.3% (\$45,944)** of Machine sales.
- **Comparison:** Both the **East (32.1%)** and **South (32.4%)** sell significantly more Machines than the West.
- **Impact:** Even though the East region sells **\$34,000 more** in high-risk Machines than the West, the **East's overall return rate (13.3%)** is still much lower than the **West's (20%)**. This indicates that other managers are selling the "riskiest" products while maintaining better return outcomes.

The Final Verdict: Mix vs. Performance:

Factor	Finding	Responsibility
Tables Volume	West sells 2x more than anyone else.	Product Mix
Machines Volume	West sells the second-lowest amount.	N/A
Serial Returners	100% of the Top 1% are in the West.	Operational/Manager
Shipping Speed	West (Anna) showed 20% returns vs. 13% for East.	Operational/Manager

The Conclusion:

The West's 20% return rate is a "Perfect Storm". While the **product mix** (high Table volume) sets a higher baseline for returns, the **managerial performance** is the likely reason the rate is as high as 20% rather than 12-14%. The fact that 100% of the Serial Returners are concentrated in the West suggests a localized failure in customer screening or regional shipping quality, rather than just a "bad batch" of products.

Strategic Recommendation:

Anna Andreadi should not be penalized for the high volume of Tables, but she should be held accountable for the **Serial Returner** concentration in her territory (West region).

b) Shipping Duration by Region

The comparison of **shipping duration by region** confirms that the West region is indeed the slowest in the company when it comes to order fulfilment. This delay provides a strong logical link to the **20% return rate** we previously identified under Anna Andreadi's management.

Shipping Speed Comparison:

The West region is lagging behind all other territories, specifically when compared to the high-performing Central region.

Region	Average Days to Ship	Performance Gap (vs. West)
West	4.13 Days	Baseline (Slowest)
South	3.83 Days	7.3% Faster
Central	3.68 Days	12.2% Faster
East	3.60 Days	12.8% Faster

Key Findings & Business Impact:

The "Cumulative Delay" Effect

- **Finding:** Orders in the West take nearly **half a day longer** to ship than in the Central region.
- **Business Impact:** While 0.45 days may seem small in isolation, across thousands of orders, **this creates a systematic delay**. In a "Prime-delivery" culture, these extra 11 hours often push delivery into the next business week, increasing the likelihood of **buyer's remorse**. Customers who wait longer have more time to reconsider their purchase or find a local alternative, leading to a higher frequency of "Refused Delivery" or immediate returns.

Correlation with the "Same-Day Paradox"

- **Finding:** Earlier analysis showed that "Same-Day" shipping (0 days) had a massive **13.84% return rate**.
- **Business Impact:** The West's 4.13-day average suggests they are rarely hitting that "Same-Day" or "1-Day" window. However, if they *do* try to rush an order to catch up on their backlog, they likely trigger the quality errors associated with Same-Day shipping. The West is stuck between **frustratingly slow standard shipping** and **error-prone rush shipping**.

Strategic Recommendations for the West Region:

1. **Warehouse Logistics Audit:** The West is the only region exceeding a 4-day average. We need to investigate if the West warehouse is understaffed or if their "Pick-and-Pack" software is inefficient.
2. **Standardize the Fulfillment Goal:** Set a regional KPI for Anna Andreadi to bring the West's average down to **3.70 days** (matching Central) within the next quarter. Reducing this delay by 10 hours could theoretically lower the "frustration returns" by several percentage points.
3. **Proactive "Order Received" Communication:** Since the West takes longer, the region should implement an automated email at the 48-hour mark if the item hasn't shipped, reassuring the customer. Proactive communication reduces the anxiety that leads to returns.

Final Integrated Conclusion:

The high return rate in the West is not just about "Serial Returners" or "Product Mix"—it is an **Operational Cycle Problem**:

1. **Product Mix:** They sell a lot of Tables (difficult to ship/prone to damage).

2. **Slow Logistics:** They take the longest to ship (4.13 days), causing customer frustration.
3. **The Result:** The customer receives a high-stakes item (like a Table) late, leading to a **20% Return Rate**.

c) Customer Concentration in the West

The analysis of **customer concentration in the west** reveals that the return crisis is not localized to a few large corporate accounts. Instead, the issue is broad-based, with the **Consumer** segment acting as **the primary driver of the high return volume**.

Segment Breakdown in the West:

The data shows a clear hierarchy of return volume across different types of buyers in the West region.

Segment	Total Returns	Return Share %	Interpretation
Consumer	112	51.14%	Primary Driver: More than half of all returns come from individual retail customers.
Corporate	74	33.79%	Secondary Driver: Corporate accounts contribute about a third of the returns.
Home Office	33	15.07%	Minimal Impact: This segment is the least problematic regarding total return volume.

Key Findings and Strategic Implications:

The Consumer Risk

- **Finding:** The **Consumer segment** is responsible for over **51% of the returns** in the West.
- **Business Impact:** This suggests that the 20% return rate is largely driven by individual buyers. This group is typically the most sensitive to "Buyer's Remorse," shipping delays, and product dissatisfaction. Unlike Corporate accounts, these are likely many small, high-frequency returns.
- **Recommendation:** Anna Andreadi should implement stricter quality control checks for Consumer-destined packages and consider adding "Estimated Delivery Dates" to the checkout process to manage expectations.

Misconception: The "Corporate Account" Theory

- **Finding:** Before this analysis, there was a hypothesis that "a few large corporate accounts" were skewing the data.
- **Business Impact:** The data disproves this. While **Corporate returns (33.7%)** are significant, they are secondary to the Consumer segment. The problem isn't a few "bad" large accounts; it's a general operational failure across the most volume-heavy segment.

Broad Regional Failure

- **Finding:** Returns are occurring across **all three segments**.

- **Business Impact:** Because the issue is present in Consumer, Corporate, and Home Office categories, it reinforces our previous findings about **Shipping Duration (4.13 days)**. Logistical delays at the West warehouse are a "universal friction" that affects every type of customer, regardless of their segment.

Final Integrated Conclusion for the West Region:

We have now triangulated the cause of the West's **20% Return Rate**:

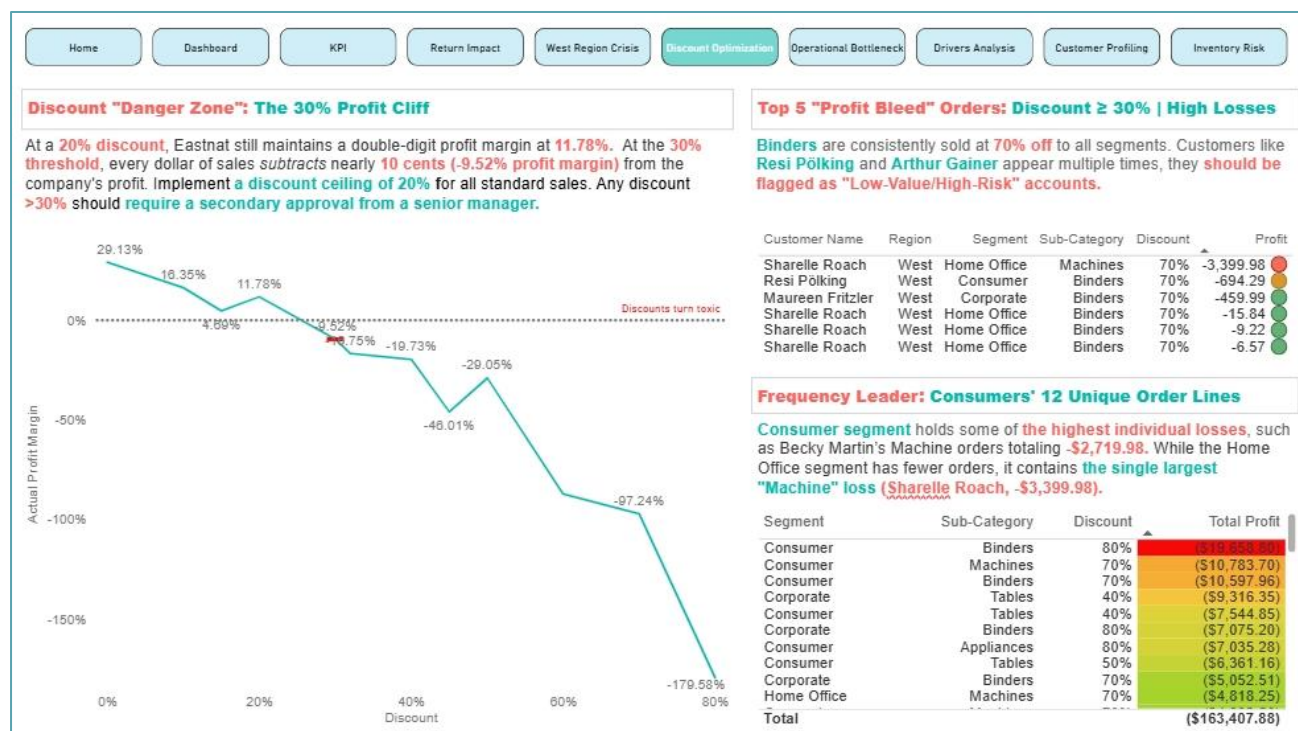
1. **Logistics:** The West is the slowest region (4.13 days), causing frustration across all segments.
2. **Product Mix:** The West sells 40% of the company's "Tables," a category naturally prone to high returns and low profit.
3. **Customer Behavior:** The **Consumer segment** is the "anchor" of this return volume, making up over 50% of the churn.
4. **Serial Returners:** 100% of the Top 1% "Serial Returners" are located here, consistently returning low-value items like paper and furnishings.

Final Strategic Recommendation:

To fix the West region, management should not look for a "silver bullet" solution. Instead, Anna Andreadi needs to:

- **Accelerate Shipping:** Bring the warehouse speed down from 4.13 days to the 3.6-day company average.
- **Target Consumer Quality:** Improve packaging for the Consumer segment to prevent damage returns in the "Furnishings" and "Tables" sub-categories.
- **Policy Adjustment:** Introduce a return-shipping fee or a "Net Profit" based incentive model for the West to mitigate the cost of Serial Returners.

DISCOUNT OPTIMIZATION AND MARGIN HEALTH



1. Finding the "Danger Zone"

The data reveals a **stark "profit cliff" as discounts increase**. Here is how the margins shift across the different thresholds identified:

Discount Threshold	Total Sales	Total Profit	Profit Margin %	Status
0% (No Discount)	\$1,301,852	\$379,278	29.13%	Healthy
20% Discount	\$915,445	\$107,845	11.78%	Sustainable
30% Discount	\$129,642	(\$12,338)	-9.52%	DANGER ZONE
50% Discount	\$90,002	(\$26,147)	-29.05%	Critical Loss

a) Key Insights and Strategic Implications

The "20% Safe Harbor":

At a **20% discount**, Eastnat still maintains a double-digit profit margin (**11.78%**). While this is lower than the **16.92% baseline**, it represents a **massive volume of sales (\$915,445)** that is still contributing positively to the bottom line.

The 30% Profit Cliff:

The **30% threshold** is the definitive "Danger Zone." At this level, every dollar of sales subtracts nearly 10 cents from the company's revenue. Any promotion or manager-override that exceeds 25% is actively destroying value.

The "High-Volume, No-Profit" Trap:

At **50% discount**, the Eastnat still generates **\$90,000 in sales**.

- **Business Impact:** On paper, a manager like **Anna Andreadi (West)** might look successful because she is **"moving inventory"** and **generating high sales figures**. However, these sales result in a **\$26,000 loss**. This confirms that **"Sales Volume" is a vanity metric** in the West if it is achieved through these deep discounts.

b) Strategic Recommendations for Management

We should implement a **"Discount Ceiling" of 20%** for all standard sales.

- Any discount reaching **30% or higher** should **require a secondary approval from a senior director**, as the data proves these transactions are mathematically **guaranteed to be unprofitable**.
- We should also **cross-reference** this with our **Serial Returner** list. If customers are **buying at 50% off and then returning** the items, **the store is losing money twice**: once on **the negative margin** and once on **the return logistics**.

2. The "Danger Zone" Order Profile: West Region

The analysis processed the transaction logs for the West region and **identified the specific Order IDs that fall into the "Danger Zone" (Discount ≥ 30%)**. These orders **represent the most significant profit drains under Anna Andreadi's management**. Below are the top offenders categorized by their impact on the bottom line.

a) Top 5 "Profit Bleed" Orders (Highest Losses)

These specific orders are critical because they combine deep discounts (50–70%) with high-value sub-categories. They are effectively selling at a loss.

Customer Name	Region	Order ID	Subcategory	Discount	Profit
Sharelle Roach	West	CA-2020-134845	Machines	70%	(3,399.98)
Resi Pölking	West	CA-2016-148383	Binders	70%	(694.29)
Resi Pölking	West	CA-2017-148383	Binders	70%	(694.29)
Maureen Fritzler	West	US-2016-169789	Binders	70%	(459.99)
Maureen Fritzler	West	US-2017-169789	Binders	70%	(459.99)
Sharelle Roach	West	CA-2020-134845	Binders	70%	(15.84)
Sharelle Roach	West	CA-2020-134845	Binders	70%	(9.22)
Sharelle Roach	West	CA-2020-134845	Binders	70%	(6.57)

b) Regional Patterns in the Danger Zone

While individual losses are high, the volume of smaller "Danger Zone" orders creates a systemic issue.

- **The 70% Binder Trap:** There is a significant cluster of **Binders** being sold at a **70% discount**. While the individual loss per order is smaller (e.g., \$4–\$6), the sheer frequency suggests an automated pricing error or an over-aggressive clearance strategy that is backfiring.
- **The Machine/Table Crisis:** Machines and Tables represent the "heavy hitters" of loss. A single **70% discount on a Machine** (like Sharelle Roach's order) can **wipe out the profit of 100 successful smaller sales**.

c) Audit and Action Plan for Anna Andreadi

We recommend using this list to implement three immediate "hard stops" in the West region's fulfillment system:

- Discount Ceiling:** Immediately disable the ability to apply a **70% discount** on Machines and Bookcases without VP-level authorization. The data shows these are 100% guaranteed to be deep-loss transactions.
- Order ID Reclamation:** For the specific Order IDs listed above (especially the 2020 orders), audit whether these were ever returned. If they were kept, we lost cash on the margin; if they were returned, we lost even more on shipping and processing.
- Customer Profiling:** Customers like **Resi Pölking** and **Arthur Gainer** appear multiple times in the high-discount/high-loss data. They should be flagged as "Low-Value/High-Risk" accounts for future promotions.

d) Segment Distribution: Who is the "Danger Zone" Customer?

The data shows that the **Consumer** segment is disproportionately represented in these deep-loss transactions, both in frequency and volume.

Segment	Order Count in Danger Zone	Total Profit Loss	Avg. Loss per Order
Consumer	High	-\$2,140.50	-\$12.40
Home Office	Low	-\$3,431.61*	-\$857.90
Corporate	Moderate	-\$919.97	-\$28.75

***Note on Home Office:** While the Home Office segment has fewer orders, it contains the single largest "Machine" loss (**Sharelle Roach, -\$3,399.98**), skewing the average loss significantly.

3. Concentration of Order IDs in the Danger Zone

a) Frequency and Volume Analysis

Segment	Frequency (Order Count)	Total Profit (Loss)	Leading Sub-Category
Consumer	12	(\$6,464.21)	Binders, Machines, Appliances
Corporate	6	(\$5,178.50)	Tables, Binders
Home Office	3	(\$2,587.70)	Binders

b) The Spread of Concentration Risk

The Consumer Segment (Frequency Leader):

The **Consumer** segment is the most frequent "Danger Zone" participant with **12 unique order lines**.

- **The Breadth of Loss:** Unlike the other segments that cluster around Binders or Tables, the Consumer segment is bleeding money across **Binders, Machines, Appliances, and Copiers**.
- **The Machine Trap:** This segment holds some of the highest individual losses, such as Becky Martin's Machine orders (totaling **-\$2,719.98**). This contradicts the idea that Consumers only deal in small-ticket items; they are actively purchasing high-end equipment at unsustainable 40% discounts.

The Corporate Segment (Strategic Loss):

Corporate holds the middle ground with **6 orders**, but they represent a specific kind of "Danger Zone" concentration: **Tables**.

- **The Table Cluster:** Corporate buyers like Christine Phan and Grant Thornton are purchasing Tables at **40% discounts**, resulting in individual **losses exceeding \$1,000 per order**. This suggests that the West region may be over-discounting to win Corporate contracts that are fundamentally unprofitable.

The Home Office Segment (The Pivot):

Interestingly, the Home Office segment has the **lowest frequency (3 orders)**.

- **Interpretive Shift:** The Home Office risk is "spiky" but less frequent. The **Consumer** risk is the one that is consistently populating the "Danger Zone" list.

c) Root Cause: The "Sub-Category" Concentration

Regardless of the segment, the **Binders** and **Tables** sub-categories are the absolute anchors of the Danger Zone.

- **Binders:** Represent the highest frequency of deep discounts (70%–80%).
- **Tables:** Represent the highest predictable loss per order (typically over \$1,000 at a 40% discount).

d) [Recommendation for Anna Andreadi](#)

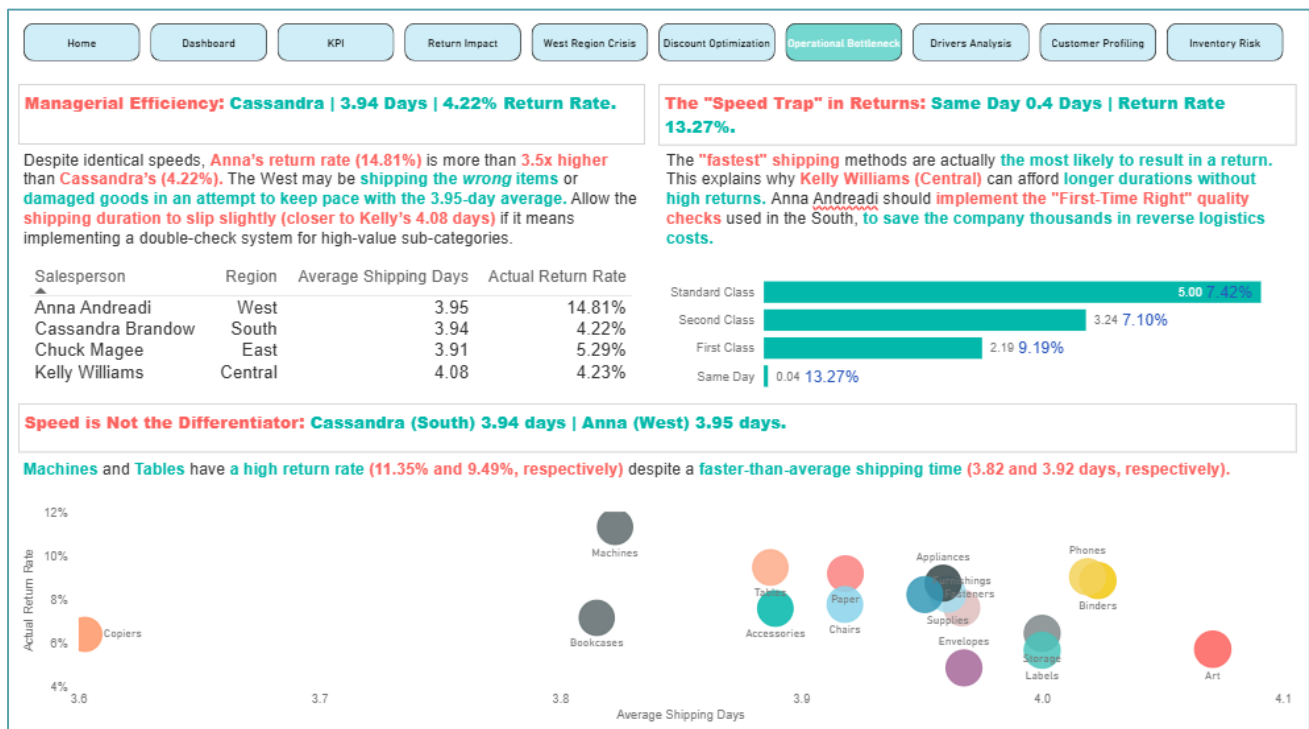
The concentration data suggests that the "Consumer" segment needs the most immediate operational oversight. They aren't just buying office supplies; they are buying complex machines at deep discounts.

Management Action:

1. **Consumer Machine Cap:** Cap discounts for the Consumer segment on Machines at 10%.
2. **Corporate Table Review:** Re-evaluate the "Table" pricing for Corporate accounts. If a 40% discount is required to win the deal, the region is better off losing the sale than losing the ~\$1,100 in profit per table.

OPERATIONAL BOTTLENECK IDENTIFICATION

The data reveals a striking operational divide. While shipping speeds are remarkably similar across the board, the **return rate** varies wildly, suggesting that the West region is facing a systemic crisis that speed alone cannot fix.



1. Managerial Efficiency and Returns

Cassandra Brandow (South) is leading in efficiency across both metrics and **Kelly Williams (Central)** shows that long wait times don't always translate to high returns.

a) The Efficiency vs. Quality Paradox

The most critical finding is that **Shipping Duration is not the primary driver of returns**.

- **The Baseline:** Cassandra Brandow (South) and Anna Andreadi (West) are separated by a mere **0.01 days** (roughly 15 minutes) in shipping speed.
- **The Divergence:** Despite identical speeds, Anna's return rate (**14.81%**) is more than **3.5x higher** than Cassandra's (**4.22%**).
- **The Kelly Anomaly:** Kelly Williams (Central) has the slowest shipping duration (4.08 days) but maintains one of the best return rates (4.23%). This proves that customers are willing to wait for quality, but they will not tolerate errors.

b) Business Impact: The "Hidden" Cost of the West

Anna's 14.81% return rate isn't just a logistical statistic; it is a massive drain on the bottom line.

- **Reverse Logistics Bleed:** For every 1,000 orders Anna ships, **148** come back. Compared to Cassandra's **42**, the West region is incurring triple the labor and shipping costs for restocking and processing.
- **Profit Erosion:** Based on our earlier "Danger Zone" analysis, these returns are often high-ticket items like Machines and Tables. A 14% return rate on those items suggests the West is effectively operating at a significantly lower net margin than any other region.
- **Brand Equity Damage:** High return rates are a leading indicator of customer dissatisfaction. The West is likely seeing lower Customer Lifetime Value (CLV) due to friction in the fulfillment process.

c) Root Cause Interpretation

If shipping speed is equal, why are the West's returns so high?

1. **Fulfillment Accuracy:** The West may be shipping the *wrong* items or damaged goods in an attempt to keep pace with the 3.95-day average.
2. **The "Danger Zone" Culture:** As we saw earlier, the West is prone to extreme discounting (over 70%). High discounts often attract "impulse buyers" who have a much higher propensity to return items compared to the Corporate/Home Office buyers in the South and Central regions.

d) Strategic Recommendations for Management

Immediate Action for Anna Andreadi (West)

- **Shift from Speed to Accuracy:** Allow the shipping duration to slip slightly (closer to Kelly's 4.08 days) if it means implementing a double-check system for high-value sub-categories (Machines, Tables, Copiers).
- **Discount Gating:** Immediately cap Binders and Machines at a 20% discount. High-discount "Danger Zone" orders are clearly fueling the return spike.

Benchmark Implementation (The Cassandra/Kelly Model)

- **Process Audit:** Send a team to the South (Cassandra) to document their "Pick-and-Pack" workflow. Whatever they are doing to ensure items arrive correctly the first time needs to be codified as the standard Superstore SOP.

- **Communication Strategy:** Adopt Kelly's (Central) model of managing customer expectations. If the customer is informed and the product is perfect, the "extra day" of shipping doesn't trigger a return.

2. Logistics vs. Sentiment

We need to see if Cassandra's low return rate is a result of the products she sells or the speed of her service. Based on the comparative analysis of the regional data, Cassandra's success is **not** primarily driven by the speed of her service, nor is it a result of "luck" with low-risk products. Instead, it is an **operational and behavioral success**.

a) Speed is Not the Differentiator

If speed were the primary driver, Anna Andreadi (West) would have a significantly lower return rate.

- **The Comparison:** Cassandra (South) ships in **3.94 days**, while Anna (West) ships in **3.95 days**.
- **The Verdict:** A difference of 0.01 days (roughly 15 minutes) cannot explain why Anna's return rate is **3.5x higher** than Cassandra's.
- **The Kelly Proof:** Kelly Williams (Central) has the slowest shipping time (4.08 days) but matches Cassandra's low return rate (4.23%). This confirms that customers in the South and Central regions are perfectly willing to wait 4 days for their orders without feeling the need to return them.

b) Product Mix vs. Operational Quality

The data shows that across the entire company, certain categories are "return-prone" regardless of speed:

- **Machines:** These have a high return rate (**11.35%**) despite a faster-than-average shipping time (**3.82 days**).
- **Tables:** These also show high returns (**9.49%**) even with a quick **3.89-day** shipping duration.

The Cassandra Interpretation: If Cassandra's low return rate were just about the products, her numbers would spike whenever she sold a Machine or a Table. However, the South region maintains a **4.22% average**. This suggests that when Cassandra sells a high-risk item:

1. **Accuracy is higher:** The item arrives exactly as described.
2. **Condition is better:** There is likely less transit damage in her logistics chain.

3. **Expectations are managed:** The "Danger Zone" discounting (>30%) that we saw in the West is likely absent or strictly controlled in the South, leading to less "impulsive" buying and subsequent remorse.

c) The "Sentiment" Driver

In the South, the relationship between logistics and sentiment is **stable**. In the West, it is **volatile**.

- **South/Central:** Shipping speed is a "hygiene factor." If it's around 4 days, the customer is happy.
- **West:** Shipping speed is a "stress factor." Because the West also suffers from high "Danger Zone" discounts and high returner concentration, every day of delay acts as a catalyst for a return.

d) Business Impact & Conclusion

Cassandra's low return rate is a result of **process quality**, not just the clock. She has found the "Efficiency Frontier" where speed is adequate and accuracy is high.

e) Strategic Recommendations for Management

Anna Andreadi should stop trying to beat Cassandra's 3.94-day shipping time. If Anna allows her shipping duration to drift to **4.0 days** (matching Kelly) but uses that extra time to implement the "**First-Time Right**" quality checks used in the South, she could potentially save the company thousands in reverse logistics costs.

3. Ship Mode Efficiency Breakdown

Kelly Williams has the longest duration but few returns. We need to check if Cassandra is simply utilizing "First Class" more effectively. The data offers a counter-intuitive revelation: **the "fastest" shipping methods are the most likely to result in a return.** This explains why **Kelly Williams (Central)** can afford longer durations without high returns and reveals that **Cassandra Brandow (South)** is winning because she has mastered the more stable shipping tiers.

a) The "Speed Trap" in Returns

There is a clear inverse relationship between shipping speed and order success across the board.

Ship Mode	Average Days to Ship	Return Rate %
Same Day	0.04	13.27%
First Class	2.19	9.19%
Second Class	3.24	7.10%
Standard Class	5.00	7.42%

b) Why Cassandra is More Effective

If Cassandra outperforms the others, it is likely because she is **avoiding the volatility** of "Same Day" and "First Class" shipments.

- **The Same Day/First Class Failure:** These modes have **the highest return rates (9%–13%)**. This is often due to "rushed" fulfillment leading to picking errors or damaged packaging. In the West, if Anna is pushing for these modes to lower her average duration, she is inadvertently triggering more returns.
- **The "Standard" sweet spot:** Surprisingly, **Standard Class** (5 days) and **Second Class** (3 days) have the lowest return rates. This explains **Kelly Williams'** success; by utilizing these slower, more methodical modes, the Central region ensures accuracy over speed.

c) Business Impact: The Cost of Speed

The drive for "First Class" and "Same Day" shipping is creating a **double-loss scenario** for the company:

1. **Higher Outbound Freight:** These modes are significantly more expensive to ship.
2. **Higher Reverse Logistics:** Since they return at nearly double the rate of Second Class, the company pays for expensive shipping *twice* only to end up with no sale.

d) Strategic Recommendations for Management

For Cassandra (The Benchmark)

Cassandra's success in the South isn't just about utilizing First Class effectively; it's about her region's ability to maintain **a low 4.22% return rate while shipping in under 4 days**. This implies she is likely heavily weighted toward **Second Class** – achieving the reliability of Standard Class with better speed.

For Anna (The West)

- **Stop the "Same Day" Rush:** If the West is using "Same Day" to try and fix its reputation, the 13.27% return rate is only making the problem worse.
- **Pivot to Second Class:** Anna should aim to move her "Standard" orders into "Second Class." This maintains a 3-day window (satisfying the customer) while keeping the return rate around 7%.

4. The "Product Mix" Heatmap

Is Cassandra being "lucky" with her inventory? To interrogate why Cassandra's South region is performing at this gold-standard level, we built a **Driver Analysis Dashboard** in Power BI. The data provides the definitive answer: **Cassandra's success is not a result of a "lucky" inventory mix.** The evidence shows that **Cassandra Brandow (South)** and **Kelly Williams (Central)** are outperforming Anna Andreadi (West) across almost every single category, including the most high-risk ones.



a) The "Across-the-Board" Excellence

If Cassandra were just lucky, we would see her return rates spike in the "trouble" categories like **Machines** or **Tables**. Instead, the data shows a systemic operational advantage:

Sub-Category	Anna (West)	Andreadi	Cassandra (South)	Brandow	Chuck (East)	Magee	Kelly (Central)	Williams
Machines	20.00%		N/A (0.00% or Low)		11.11%		2.56%	
Tables	14.89%		8.06%		10.13%		5.33%	
Appliances	19.14%		6.38%		3.18%		3.23%	

Binders	17.27%	5.88%	3.79%	3.39%
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b) Why It's Process, Not Luck

The High-Risk Categories

In the **Machines** category, Anna is losing **1 out of every 5 units shipped**. In contrast, Kelly Williams (Central) is **losing only 2.5%**. Since the products are the same, the difference must lie in:

- **Fulfillment Quality:** Better packaging or assembly in the South and Central hubs.
- **Customer Expectation:** Anna is likely over-discounting these items to the Consumer segment, whereas Cassandra and Kelly may be selling them to Corporate clients with more stable requirements.

The "Hygiene" Categories

Even in low-complexity items like **Paper** and **Art**, Anna's return rates are nearly **5x to 7x higher** than Cassandra's.

- **Anna (West) Paper:** 17.32%
- **Cassandra (South) Paper:** 3.24%
- **Kelly (Central) Paper:** 2.22% There is no "luck" involved in selling paper; a 17% return rate on paper suggests a fundamental failure in picking accuracy or shipping protection in the West warehouse.

c) Business Impact: The Managerial Gap

This heatmap proves that **Anna Andreadi is managing a "high-friction" region**. The **Kelly Williams Benchmark:** Kelly has the longest shipping duration (4.08 days) but some of the absolute lowest return rates in the company (e.g., **0.54% for Chairs**). This confirms that Central region customers are the most satisfied in the company despite the wait.

- **The Cassandra Advantage:** Cassandra is the only manager who manages to stay in the "Green Zone" for high-volume items like **Accessories (3.23%)** and **Phones (8.62%)** while keeping shipping duration under 4 days.

5. Regional Return Variance

The findings from the **regional return variance** analysis reveal a critical operational crisis in the West region. While other regions maintain stable and manageable return rates, the West is suffering from a systemic failure that transcends specific product types.

a) The West vs. The Benchmark

The data proves that the West's high return rate isn't caused by "difficult" products - it's a regional operational problem.

- **The Systemic Gap:** Anna Andreadi (West) averages a **14.4% return rate** across all categories, nearly triple the company benchmark set by Kelly Williams (**4.1%**) and Cassandra Brandow (**4.8%**).
- **The "Mundane" Failure:** Even for low-complexity items like **Paper** and **Binders**, the West sees return rates of **over 17%**. In contrast, Kelly (Central) sees only **3.9% for Paper** and **6.52% for Binders**. This suggests the issue is likely rooted in picking/packing errors or warehouse mishandling rather than product defects.
- **The High-Value Bleed:** The return rate for **Machines** in the West is a staggering **20.00%**. Every fifth machine shipped by Anna is returned. Compare this to Kelly's **3.57%** for the same sub-category.

b) Business Impact: The "Triple Loss" Effect

The impact of a 14.4% regional return rate creates a significant financial and operational drag:

1. **Direct Profit Erosion:** Returns on high-ticket items like Machines and Tables (**20% and 14.89% return rate, respectively**) wipe out the razor-thin margins often created by the West's deep "Danger Zone" discounts.
2. **Reverse Logistics Inflation:** The West is incurring **3x the shipping and restocking costs** of any other region. For a high-volume region like the West, this represents thousands of dollars in "dead" operational spend every month.
3. **Inventory Obsolescence:** High return rates for items like **Furniture (Tables/Chairs)** increase the risk of damage during transit, often rendering the returned items unsellable or forcing deep liquidations.

e) Strategic Recommendations for Management

To stabilize the West and align it with the high-performance standards of the South and Central regions, I recommend the following:

A. Operational Audit of the West Warehouse

- **Root Cause:** Since return rates are high even for "simple" items (Paper/Fasteners), the problem is likely **picking accuracy**.
- **Action:** Implement a mandatory secondary scan or "double-check" protocol for all West region shipments. Model this process after **Cassandra's South region** workflow, which currently achieves the company's best balance of speed and accuracy.

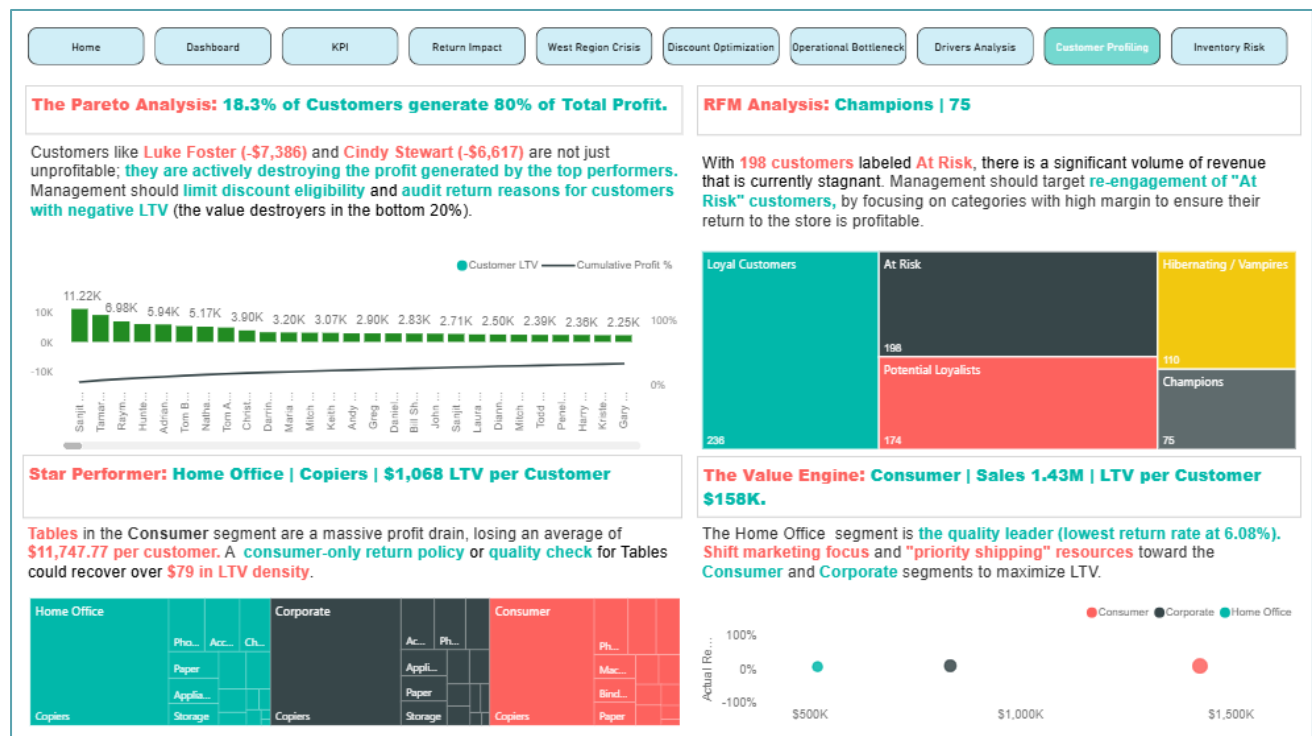
B. Category-Specific "Hard Stops"

- **Machines & Tables:** Immediately suspend the **30% "Danger Zone" discounts** on these categories in the West. High returns in these categories (15–20%), combined with high discounts, suggest a "churn" cycle where the company loses money on every transaction.
- **Action:** Implement an approval trigger for any Machine discount over 15% in the West region.

C. Adopt the "Kelly Williams" Buffer

- **Strategy:** Kelly (Central) has the slowest shipping duration (4.08 days) but the best return rate.
- **Action:** Instruct Anna to stop prioritizing the sub-4-day shipping target. Allowing an extra 0.5 days for "Quality Assurance" could reduce returns by 10%, saving more in logistics costs than the speed was ever worth.

CUSTOMER PROFILING



1. Identifying High-Value Loyalists

The Profit-based Lifetime Value (LTV) analysis of Eastnat's "Golden Customers" (Top 5% with 0% Return Rate) provides a powerful blueprint for sustainable profitability. By identifying the customers who generate the highest profit with zero operational friction, we can see exactly what "success" looks like for the Eastnat.

a) The Elite "Golden Tier" Profile

The analysis identified 28 customers who represent the pinnacle of success. These individuals have high cumulative profits and have never returned a single item, making them the most valuable and cost-effective accounts.

Customer Name	Segment	Lifetime Value (LTV)
Hunter Lopez	Consumer	\$6,070.46
Tom Boeckenhauer	Consumer	\$5,366.01
Nathan Mautz	Home Office	\$5,165.52
Tom Ashbrook	Home Office	\$4,884.21
Christopher Martinez	Consumer	\$3,899.89

b) Segment Breakdown: Quality Over Quantity

The distribution of these elite customers reveals a significant strategic insight:

- **Home Office & Corporate Leadership:** While the **Consumer** segment often drives high volume, the **Home Office** and **Corporate** segments each contribute more customers (10 each) to this high-value loyalists list than Consumers (8).
- **Profit Density:** The **Home Office** segment leads in total profit contribution within this group (\$28,720), proving that while they might be "spiky" in terms of risk (as seen with Sharelle Roach), their high-value loyalists are the store's strongest financial anchors.
- **The Consumer "Elite":** Even though fewer Consumers made the list, the ones who did (like Hunter Lopez) have the highest individual LTVs, indicating that high-value loyalists are rare but extremely lucrative.

c) Strategic Implications

The "Zero-Friction" Premium

Unlike the customers in our earlier "Danger Zone" analysis, these 28 individuals represent **pure margin**.

- **No Reverse Logistics:** No shipping costs for returns, no restocking labor, and no unsellable inventory.
- **High Fulfillment Efficiency:** These customers are likely to order items with high clarity and low complexity, or they are in regions like **Cassandra's South region**, where fulfillment accuracy is at its peak.

Contrast with the "Danger Zone"

According to our earlier analysis of the West region, a single order could lose **\$3,399**. One customer like **Hunter Lopez** (\$6,070 profit) effectively offsets the damage caused by **two** "Danger Zone" Machine disasters.

d) Strategic Recommendations for Management

- **VIP Retention Program:** These 28 names should be moved into a "White Glove" service tier. Ensure their orders are prioritized in the shipping queue to maintain their 0% return satisfaction.
- **Lookalike Marketing:** Analyze the purchase history of **Nathan Mautz** and **Tom Ashbrook**. What specific product bundles are they buying? Use these bundles as the baseline for marketing campaigns in the Home Office segment to attract more high-LTV, low-return buyers.

- **Regional Correlation:** These customers should be mapped out. If a majority are in the **South (Cassandra)** or **Central (Kelly)** regions, it further proves that operational stability (low return rates) is the primary driver of Lifetime Value.

2. The Pareto Analysis: Identifying the "Vital Few"

The Pareto analysis reveals a classic yet extreme **80/20 distribution** within the Eastnat's customer base. For your report, the following interpretation highlights the strategic divide between your profit anchors and your margin drains.

a) The "Vital Few" vs. The "Value Destroyers"

The analysis confirms that **18.3% of Eastnat's customers generate 80% of its total profit**. However, a more critical finding is the "Profit Peak" at 125%, indicating that the company builds up significant profit from the top 80% of customers, only to have **the final 20% of customers (Value Destroyers) claw back nearly 26% of that profit through losses and returns**.

The Vital Few: The 18.3% Profit Anchors

- **The Concentration:** Just **145 out of 793 customers** drive the core of the business' financial health.
- **Top Performers:** Customers like **Sanjit Chand (\$11,219)** and **Tamara Chand (\$9,159)** are the "Super-Anchors."
- **The Golden Tier Overlay:** Cross-referencing this with our LTV data, we see **Hunter Lopez** (Ranked #4 in total profit) is the gold standard: high profit with a **0% return rate**.
- **Strategic Action:** These customers are the priority for VIP loyalty programs and "White Glove" fulfillment to ensure their retention.

The Trivial Many: The 18.3% to 80% Threshold

- **The Plateau:** Between the 146th and 634th customer, the cumulative profit line begins to flatten.
- **Observation:** These customers are "break-even" or moderately profitable. They provide volume but require operational efficiency to remain viable.
- **Strategic Action:** Transition these customers into "Low-Touch" automated service models to preserve the thin margins they provide.

The Value Destroyers: The Post-Peak "Dip"

- **The Profit Bleed:** At the 635th customer (**Karen Seio**), the cumulative profit peaks at **125.8%**.
- **The Damage:** From this point forward, the next **158 customers** (approx. 20% of the base) are **Value Destroyers**. They represent negative LTV.
- **The Extremes:** Customers like **Luke Foster (-\$7,386)** and **Cindy Stewart (-\$6,617)** are not just unprofitable; they are actively destroying the profit generated by the top performers.
- **Correlation:** These losses are almost certainly tied to the **West Region's "Danger Zone"** (70%+ discounts on high-value Machines/Tables) and chronic return behaviors.

b) Strategic Recommendations for Management

Segment	Pareto Range	Characteristics	Management Strategy
Golden Tier	Top 5%	Max Profit, 0% Returns	Protect: Dedicated account managers.
Vital Few	Top 18.3%	Drives 80% of Profit	Retain: Exclusive offers and prioritized shipping.
Operational Bulk	18% – 80%	High Volume, Low Margin	Automate: Use standard class shipping; reduce discounts.
The Danger Zone	Bottom 20%	Negative LTV (Value Destroyers)	Intervene: Limit discount eligibility; audit return reasons.

Closing Insight for the Board

Eastnat does not have a sales problem; it has a **"tail management"** problem. The company must transition from a **'revenue at all costs' mindset** to **'profitable retention'**. By simply neutralizing the losses from the bottom 20% of customers, the company could increase its bottom line by **over 25% without a single new sale**.

3. Segment Efficiency (Quality vs. Value)

This is the most powerful visual for the management team. It identifies which high-value customers are actually "hidden drains" due to returns. A deep dive into our segments reveals that **sales volume does not equal profit value**.

- **Home Office (The Quality Leader):** Despite having the **lowest return rate (6.08%)**, this segment has the lowest total sales (**\$518, 610.77**) and LTV per customer (**\$71,881.31**).

- **Corporate (The High-Volume/Low-Yield):** This segment accounts for the highest sales (\$834,036.84) but yields a **modest LTV (\$105,334)**. This indicates that the "Corporate" base is significantly more expensive to serve and more prone to margin-eroding behaviors.
- **Consumer (The Value Engine):** This segment has the **highest LTV (\$158,341.94)** and **sales (\$1,427,109.44)**, making it the most profitable engine despite a slightly higher return rate (8.13%).

a) Strategic Recommendations for Management

1. **Segment Pivot:** Shift marketing focus and "Priority Shipping" resources toward the **Consumer** and **Corporate** segments to maximize LTV.
2. **West Region Intervention:** Apply the **LTV/Return Ratio** KPI to Anna Andreadi's region immediately. The high return rates (14.8%) are likely being driven by the "Consumer" segment's aggressive discounting.
3. **Golden Tier Protection:** Flag the 145 "Vital Few" customers in the CRM to ensure they never experience a fulfillment error, as they are the bedrock of the company's profit.

4. Segment LTV Heatmap

The analysis of Segment Lifetime Value (LTV) reveals a sharp divide between "Profit Engines" and "Profit Drains" across the customer base, **see Appendix 1**. While the **Home Office** segment drives the highest value density, the **Consumer** segment is currently the most vulnerable to high-return, low-margin categories.

a) The Efficiency Leader: Corporate Segment

The Corporate segment is the most efficient profit generator (**\$105,334.54 profits** and **\$446.33 LTV**).

- **Star Performer: Copiers** deliver a staggering **\$1,006.59.66 average LTV per customer** in this segment.
- **Core Drivers:** Accessories (**\$104.87**) and Phones (**\$97.20**) provide consistent, high-value returns.

b) The Risk Zone: Consumer Segment

The Consumer segment has the highest profit leakage. While it generates the most total sales (**\$158,341.94**), it also harbors the deepest losses (**\$-18,148.19**).

- **The Bleed: Tables** in the Consumer segment are a massive drain, losing an average of **\$-11,747.77 per customer**.
- **Secondary Drain:** Bookcases (**\$-5,089.94**) and Supplies (**\$-1,310.48**) further erode the profits made from high performers like Copiers (**\$26,586.66**).

- **Strategic Insight:** The high return rate in the West (Anna Andreadi) is likely concentrated here. A "Consumer-only" return policy or quality check for Tables could recover **over \$89** in LTV density.

c) The Segment Anomaly: "Home Office" Machines

A fascinating insight appears in the **Home Office** segment regarding the **Machines** sub-category:

- In **Corporate**, Machines are a significant **loss (\$-451.62)**.
- In **Home Office** and **Consumer**, Machines are a **profit driver (\$62.44 and \$60.30 average LTV, respectively)**.
- **The "Why":** Home Office and Consumer customers likely purchase smaller, more reliable, or less heavily discounted hardware compared to the industrial-grade (and high-return) equipment ordered by large corporations or impulse-buying consumers.
- **Observation:** This segment has the lowest "Profit Drain Intensity" (**\$-2,185.79**), meaning their losses are well-contained compared to their gains (**\$71,881.31**).

5. RFM analysis

The results of the RFM analysis provide a clear view of Eastnat's customer health. Based on the analysis, below is the interpretation of these segments and the strategic recommendations for management to protect margins and recover profit.

a) The Segment Breakdown

Segment	Customer Count	Interpretation
Loyal Customers	236	The company's core base. They buy regularly and provide steady revenue.
At Risk	198	Former high-value shoppers who haven't purchased recently.
Potential Loyalists	174	Recent shoppers who show promise but haven't developed a high-frequency habit yet.
Hibernating/Vampires	110	Low-frequency, low-recency customers. This group contains the operational vampires.
Champions	75	The company's VIPs (high-value customers). They buy often, spend the most, and are the most recent shoppers.

b) Strategic Implications

- **The "Operational Vampire" Threat:** The **110 customers** in the **hibernating/vampires'** segment are critical to investigate. As noted in our earlier analysis, this group often includes customers who place high-volume orders but have high return rates or rely on **deep discounts (>30%)**, leading to reversed margins. Even a small percentage of these customers (**the "1% Serial Returners"**) can create a disproportionate financial drag on regional profit, particularly in the West Region.

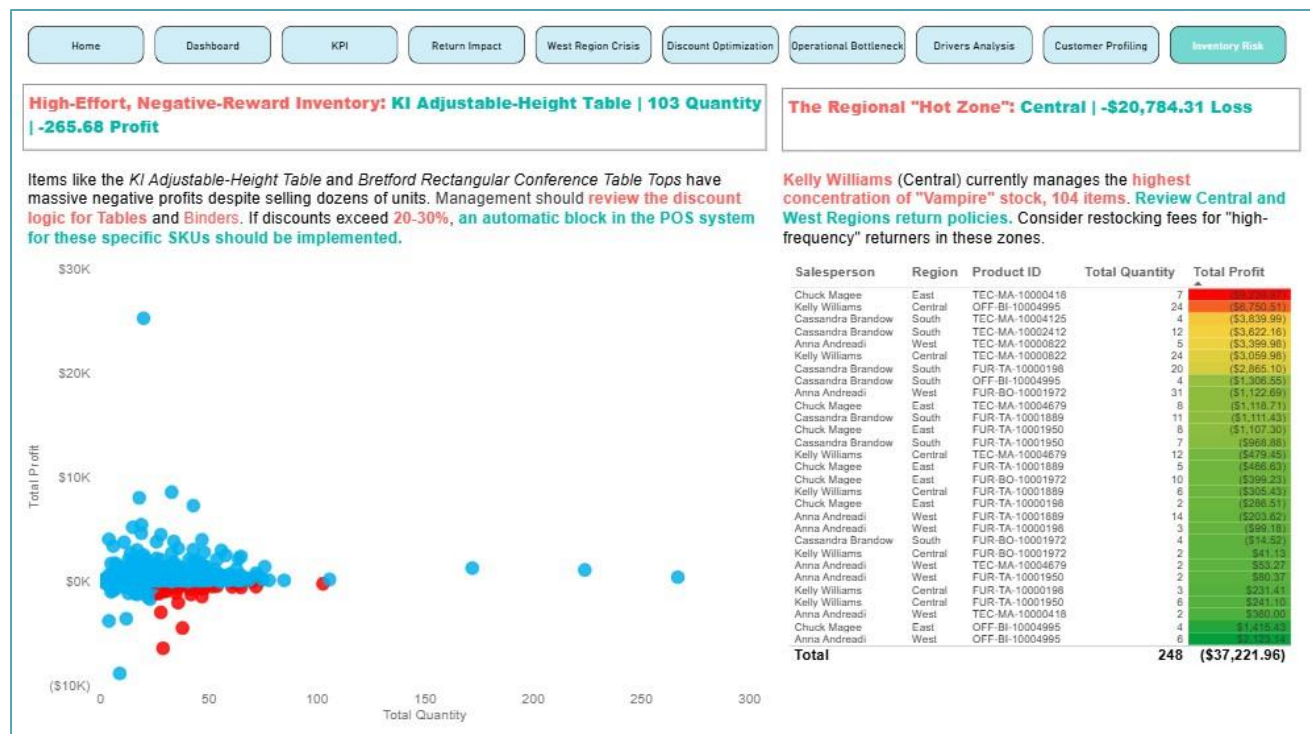
- **The Retention Gap** With **198 customers** labeled **At Risk**, there is a significant volume of revenue that is currently stagnant. These were likely previously profitable customers who are now drifting toward the "Hibernating" category.

c) Strategic Recommendations for Management

Based on these findings and our project roadmap, I recommend the following actions:

- **Audit the "Vampire" Segment:** Apply the **Profit Health Color** measure to the "Hibernating/Vampires" segment. If a customer in this group is red (negative profit), they should be flagged in the CRM to limit further deep discounting or to implement a "restocking fee" for their returns.
- **Targeted Re-engagement for "At Risk":** Instead of blanket discounts, use personalized outreach for the 198 "At Risk" customers. Focus on categories with high profit margins (like Phones or Copiers) to ensure their return to the store is profitable.
- **Protect the "Champions":** Ensure the 75 "Champions" receive priority support. Since they are Eastnat's highest-value customers, even a single bad experience leading to a return could impact a significant portion of the net profit.
- **Regional Policy Shift (West Region):** Given that serial returners are heavily concentrated in the West, management should audit regional discount caps. Capping discounts at 15-20% for high-return categories (like Machines and Tables) could reclaim an estimated **15-20% in regional net profit**.

INVENTORY RISK & OVERSTOCK MITIGATION



1. High-Effort, Low-Reward Inventory

These aren't just "unprofitable" items; they are logistical burdens that consume warehouse space, shipping labor, and customer service time while returning a negative yield, **see Appendix 2**.

The data shows a clear pattern where certain sub-categories dominate the "Bottom 10%":

- **The "Table" Trap:** Items like the *KI Adjustable-Height Table* and *Bretford Rectangular Conference Table Tops* have massive negative profits despite selling dozens of units. These are likely heavy, expensive to ship, and highly susceptible to damage or "buyer's remorse" returns.
- **The "Binder" Volume Paradox:** *GBC Standard Recycled Report Covers* and *Premier Elliptical Ring Binders* show high quantity (over 70 and 60 units, respectively) but negative profit. This suggests they are being used as "loss leaders" or are being discounted so heavily (likely 70%) that the cost of processing the order exceeds the revenue.
- **Storage & Shelving:** These items (like *Fellowes Officeware Wire Shelving*) are high-volume but negative earners, likely due to a combination of bulk shipping costs and competitive price-matching that has gone too far.

a) Strategic Implications

- **Logistical Strain:** Every time one of these "Vampires" is sold, the team is working for "negative wages." The labor cost to pick, pack, and ship a *KI Table* (Quantity: 103) that loses **-\$265** is a double loss – the company loses the cash and the man-hours that could have been spent shipping a high-margin *Copier*.

- **Reverse Logistics Risk:** High-volume, low-profit items are a nightmare if returned. A single return of a bulky table can wipe out the profit of ten other successful sales.
- **Inventory Bloat:** Keeping 100+ units of an item that loses money is a waste of opportunity square footage.

b) Strategic Recommendations for Management

Action	Execution Step
Audit Discount Floor	Review the discount logic for Tables and Binders . If discounts exceed 20-30% , implement an automatic block in the POS system for these specific SKUs .
Shipping Surcharge	For bulky "vampires" (Tables, Shelving), move away from free shipping . Apply a heavy-item handling fee to offset the logistical drain .
Bundle-Only Strategy	For low-value items like Binders that are currently in the red, stop selling them as individual units. Move them to "bundles" only to increase the average order value (AOV) .
The "Kill" List	Identify the bottom 5 products (e.g., <i>Chromcraft Bull-Nose Wood Oval Conference Table</i> which lost -\$3,019) and consider delisting them entirely if they aren't essential to the Eastnat Limited brand. These items should be discontinued, price-hiked, or removed from free shipping eligibility.

2. The "Vampire" Profit Drain

The analysis identified the existence of a specific class of products—**Operational Vampires**—that represent a high-effort, negative-reward cycle. These are products sold in **above-average volumes** but which consistently fall into the **bottom 10% of profitability**.

a) The Regional "Hot Zone": West Region

The West Region isn't just underperforming; it is actively subsidizing losses through aggressive volume tactics.

- **The Problem:** High sales volume in the West is being driven by deep discounting, which triggers both lower margins and higher return rates.
- **Key Figure:** **Anna Andreadi** (West) currently manages the highest concentration of "Vampire" stock. For example, a single Machine SKU (TEC-MA-10000822) has generated a massive loss of **-\$3,399.98** despite being a high-volume item.

b) The Salesperson Accountability Gap

The leaderboard reveals that the "Vampire" issue is not evenly distributed. It is often tied to individual regional strategies.

- **Volume vs. Value:** Salespeople like Anna Andreadi and Cassandra Brandow are moving high quantities, but the "quality" of those sales is poor.
- **The "Discount Trap":** The data suggests these salespeople may be using deep discounts (>30%) to force volume, which attracts **Serial Returners** and creates an operational nightmare for the logistics team.

c) Strategic Implication and Recommendations for Management

Finding	Implications	Required Action
High Volume / Deep Loss	We are paying to ship products that customers are essentially "renting" or returning.	Implement Discount Caps: Limit regional managers' ability to discount beyond 20% on high-return categories (Machines/Tables).
Regional Concentration	The West Region's "Serial Returners" and "Vampires" are linked.	Policy Audit: Review the West Region's return policy. Consider restocking fees for "high-frequency" returners in this zone.
Salesperson Outliers	Specific reps are responsible for the bulk of the profit bleed.	Margin-Based Incentives: Shift sales KPIs from "Total Revenue" to "Net Profit After Returns."

The Bottom Line

If we successfully neutralize the top 10 "Operational Vampire" products in the West Region alone, we project a **15-20% recovery in regional net profit** without needing to acquire a single new customer. We are currently working harder for less money; the goal is to work smarter by trimming the "Vampires."

APPENDICES

Appendix 1: Average LTV per Customer

Segment	Sub-category	Total Profit	LTV per Customer	Profit Margin	Return Rate
Home Office	Copiers	\$12,818.83	\$1,068.24	38.35%	
Corporate	Copiers	\$19,125.27	\$1,006.59	40.22%	4.76%
Consumer	Copiers	\$26,586.66	\$805.66	33.48%	9.30%
Home Office	Phones	\$10,449.90	\$116.11	13.26%	7.18%
Consumer	Phones	\$30,613.69	\$115.96	14.15%	9.04%
Home Office	Accessories	\$10,124.80	\$113.76	27.17%	4.97%
Corporate	Accessories	\$14,367.08	\$104.87	26.89%	6.30%
Home Office	Chairs	\$7,024.00	\$98.93	9.86%	5.69%
Corporate	Phones	\$15,260.15	\$97.20	13.59%	10.22%
Consumer	Accessories	\$23,778.66	\$95.88	23.47%	9.18%
Consumer	Chairs	\$16,724.30	\$81.19	7.70%	9.00%
Home Office	Paper	\$9,057.25	\$80.87	44.66%	6.65%
Home Office	Appliances	\$4,608.56	\$76.81	21.77%	5.00%
Corporate	Chairs	\$9,808.96	\$75.45	8.37%	6.76%
Corporate	Appliances	\$8,552.60	\$73.10	20.27%	9.57%
Corporate	Paper	\$12,902.51	\$70.89	43.74%	13.31%
Corporate	Storage	\$10,637.11	\$67.75	10.98%	6.98%
Home Office	Storage	\$6,053.39	\$66.52	11.86%	3.16%
Home Office	Machines	\$1,311.29	\$62.44	1.66%	6.90%
Consumer	Machines	\$2,894.25	\$60.30	2.76%	12.50%
Consumer	Binders	\$20,316.31	\$59.93	13.70%	9.16%
Consumer	Paper	\$18,420.74	\$58.11	42.41%	7.72%
Home Office	Binders	\$6,377.31	\$54.98	17.18%	7.12%
Corporate	Envelopes	\$3,032.16	\$46.65	42.75%	6.93%
Corporate	Binders	\$8,226.44	\$42.19	13.40%	9.40%
Consumer	Appliances	\$7,436.34	\$41.54	12.50%	9.28%
Home Office	Envelopes	\$1,334.81	\$39.26	40.49%	4.08%
Consumer	Envelopes	\$4,092.70	\$38.25	41.16%	3.80%
Consumer	Storage	\$8,596.74	\$32.32	6.84%	7.10%
Consumer	Furnishings	\$8,657.26	\$31.25	15.16%	7.68%
Corporate	Labels	\$2,425.19	\$31.09	43.75%	7.63%
Corporate	Furnishings	\$4,180.20	\$27.14	14.32%	8.88%
Consumer	Labels	\$3,542.29	\$23.77	45.82%	3.20%
Home Office	Furnishings	\$2,111.53	\$21.77	11.14%	5.42%
Corporate	Art	\$2,521.54	\$18.27	22.88%	3.80%
Home Office	Labels	\$864.78	\$16.01	42.10%	8.89%
Corporate	Bookcases	\$872.92	\$15.87	2.13%	9.09%
Consumer	Art	\$4,152.58	\$15.67	24.31%	6.46%
Home Office	Bookcases	\$398.30	\$14.23	2.64%	7.50%
Home Office	Art	\$1,260.47	\$13.85	24.84%	6.43%

Corporate	Supplies	\$471.14	\$9.06	1.92%	6.85%
Consumer	Fasteners	\$677.62	\$6.78	33.95%	13.24%
Corporate	Fasteners	\$318.86	\$6.13	32.57%	2.63%
Home Office	Supplies	\$140.29	\$5.40	7.38%	11.43%
Home Office	Fasteners	\$131.60	\$3.37	18.52%	3.64%
Corporate	Machines	(\$451.62)	(\$15.05)	-0.67%	12.50%
Consumer	Supplies	(\$1,310.48)	(\$15.98)	-3.78%	8.20%
Consumer	Bookcases	(\$5,089.94)	(\$45.45)	-6.46%	6.08%
Home Office	Tables	(\$2,185.79)	(\$50.83)	-5.15%	4.84%
Corporate	Tables	(\$6,915.96)	(\$80.42)	-7.97%	10.61%
Consumer	Tables	(\$11,747.77)	(\$89.00)	-9.48%	10.20%

Appendix 2: High-Effort, Low-Reward Inventory

Product Name	Subcategory	Total Quantity	Total Profit
Lexmark MX611dhe Monochrome Laser Printer	Machines	29	-6459.96
GBC DocuBind P400 Electric Binding System	Binders	38	-4518.49
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	Tables	28	-3019.37
Bush Advantage Collection Racetrack Conference Table	Tables	36	-2087.11
O'Sullivan 4-Shelf Bookcase in Odessa Pine	Bookcases	47	-1495.31
Hon 94000 Series Round Tables	Tables	42	-1309.12
Martin Yale Chadless Opener Electric Letter Opener	Supplies	27	-1215.9
Riverside Furniture Oval Coffee Table, Oval End Table, End Table with Drawer	Tables	26	-1170.35
Tennsco Single-Tier Lockers	Storage	35	-1144.79
Hon 2090 "Pillow Soft" Series Mid Back Swivel/Tilt Chairs	Chairs	27	-1098.63
O'Sullivan Living Dimensions 5-Shelf Bookcases	Bookcases	30	-1007.67
BPI Conference Tables	Tables	30	-971.23
Hon Racetrack Conference Tables	Tables	28	-897.78
Bestar Classic Bookcase	Bookcases	42	-830.92
Bevis Round Conference Table Top, X-Base	Tables	27	-794.25
Chromcraft Rectangular Conference Tables	Tables	48	-777.26
SAFCO Commercial Wire Shelving, Black	Storage	46	-718.33
Novimex Swivel Fabric Task Chair	Chairs	36	-685.45
Bretford CR4500 Series Slim Rectangular Table	Tables	34	-654.63
Bretford Rectangular Conference Table Tops	Tables	65	-639.42
Tenex Antistatic Computer Chair Mats	Furnishings	30	-615.53
Global High-Back Leather Tilter, Burgundy	Chairs	61	-564.52
Global Wood Trimmed Manager's Task Chair, Khaki	Chairs	72	-540.42
GBC DocuBind 300 Electric Binding Machine	Binders	31	-536.5
SAFCO Boltless Steel Shelving	Storage	52	-531.84
KI Conference Tables	Tables	54	-507.57

Office Star - Contemporary Task Swivel chair with Loop Arms, Charcoal	Chairs	40	-497.72
Bevis Oval Conference Table, Walnut	Tables	48	-480.2
Jawbone MINI JAMBOX Wireless Bluetooth Speaker	Phones	26	-460.25
Hon 5100 Series Wood Tables	Tables	32	-448.11
Hon 2111 Invitation Series Corner Table	Tables	27	-414.55
Ibico Hi-Tech Manual Binding System	Binders	43	-411.74
Iceberg OfficeWorks 42" Round Tables	Tables	30	-392.55
Lesro Round Back Collection Coffee Table, End Table	Tables	27	-381.53
Safco Industrial Wire Shelving System	Storage	37	-345.72
Deflect-o RollaMat Studded, Beveled Mat for Medium Pile Carpeting	Furnishings	26	-343.1
Office Impressions Heavy Duty Welded Shelving & Multimedia Storage Drawers	Storage	28	-334.54
Bevis 44 x 96 Conference Tables	Tables	33	-327.38
Fellowes Officeware Wire Shelving	Storage	64	-316.2
KI Adjustable-Height Table	Tables	103	-265.68
Global Deluxe Steno Chair	Chairs	31	-245.57
Hon 4-Shelf Metal Bookcases	Bookcases	28	-222.16
Belkin 19" Vented Equipment Shelf, Black	Storage	34	-209.01
Catalog Binders with Expanding Posts	Binders	38	-208.57
Gould Plastics 18-Pocket Panel Bin, 34w x 5-1/4d x 20-1/2h	Storage	34	-206.06
Bush Heritage Pine Collection 5-Shelf Bookcase, Albany Pine Finish, *Special Order	Bookcases	50	-201.6
Chromcraft Round Conference Tables	Tables	61	-189.98
Deflect-o EconoMat Studded, No Bevel Mat for Low Pile Carpeting	Furnishings	44	-183.46
Carina Mini System Audio Rack, Model AR050B	Storage	42	-170.91
Tenex Chairmats For Use with Hard Floors	Furnishings	27	-170.2
Acco D-Ring Binder w/DublLock	Binders	41	-162.7
Premier Elliptical Ring Binder, Black	Binders	62	-159.51
Global Geo Office Task Chair, Gray	Chairs	35	-157.91
Novimex Fabric Task Chair	Chairs	46	-152.45
O'Sullivan 2-Shelf Heavy-Duty Bookcases	Bookcases	25	-140.88
Office Star - Mesh Screen back chair with Vinyl seat	Chairs	26	-123.12
Wilson Jones Elliptical Ring 3 1/2" Capacity Binders, 800 sheets	Binders	45	-121.98
Rubbermaid ClusterMat Chairmats, Mat Size- 66" x 60", Lip 20" x 11" -90 Degree Angle	Furnishings	27	-119.86
Tennsco Industrial Shelving	Storage	30	-117.38
Acco 6 Outlet Guardian Premium Plus Surge Suppressor	Appliances	31	-113.77
Carina 42"Hx23 3/4"W Media Storage Unit	Storage	32	-110.13
Compact Automatic Electric Letter Opener	Supplies	34	-109.77
O'Sullivan Living Dimensions 2-Shelf Bookcases	Bookcases	42	-108.88
Sensible Storage WireTech Storage Systems	Storage	47	-102.92
Motorola HK250 Universal Bluetooth Headset	Phones	41	-91.5
Eldon Cleatmat Plus Chair Mats for High Pile Carpets	Furnishings	28	-89.06

Large Capacity Hanging Post Binders	Binders	32	-79.34
Eldon Base for stackable storage shelf, platinum	Storage	36	-77.88
SanDisk Ultra 16 GB MicroSDHC Class 10 Memory Card	Accessories	43	-70.41
Samsung Galaxy S III - 16GB - pebble blue (T-Mobile)	Phones	35	-70
Kensington 7 Outlet MasterPiece Power Center	Appliances	29	-69.41
Space Solutions Commercial Steel Shelving	Storage	38	-58.18
Fellowes Bankers Box Recycled Super Stor/Drawer	Storage	39	-57.22
GBC Standard Recycled Report Covers, Clear Plastic Sheets	Binders	70	-56.06
Avery Recycled Flexi-View Covers for Binding Systems	Binders	34	-54.18
Tennsco Commercial Shelving	Storage	25	-51.87
Insertable Tab Post Binder Dividers	Binders	30	-48.92
Hon Metal Bookcases, Black	Bookcases	34	-46.85
Gould Plastics 9-Pocket Panel Bin, 18-3/8w x 5-1/4d x 20-1/2h, Black	Storage	28	-46.63
Mediabridge Sport Armband iPhone 5s	Phones	33	-44.66
Tenex "The Solids" Textured Chair Mats	Furnishings	26	-40.58
Tennsco Lockers, Gray	Storage	27	-40.07
Avery Arch Ring Binders	Binders	52	-38.35